

BFS Traversal

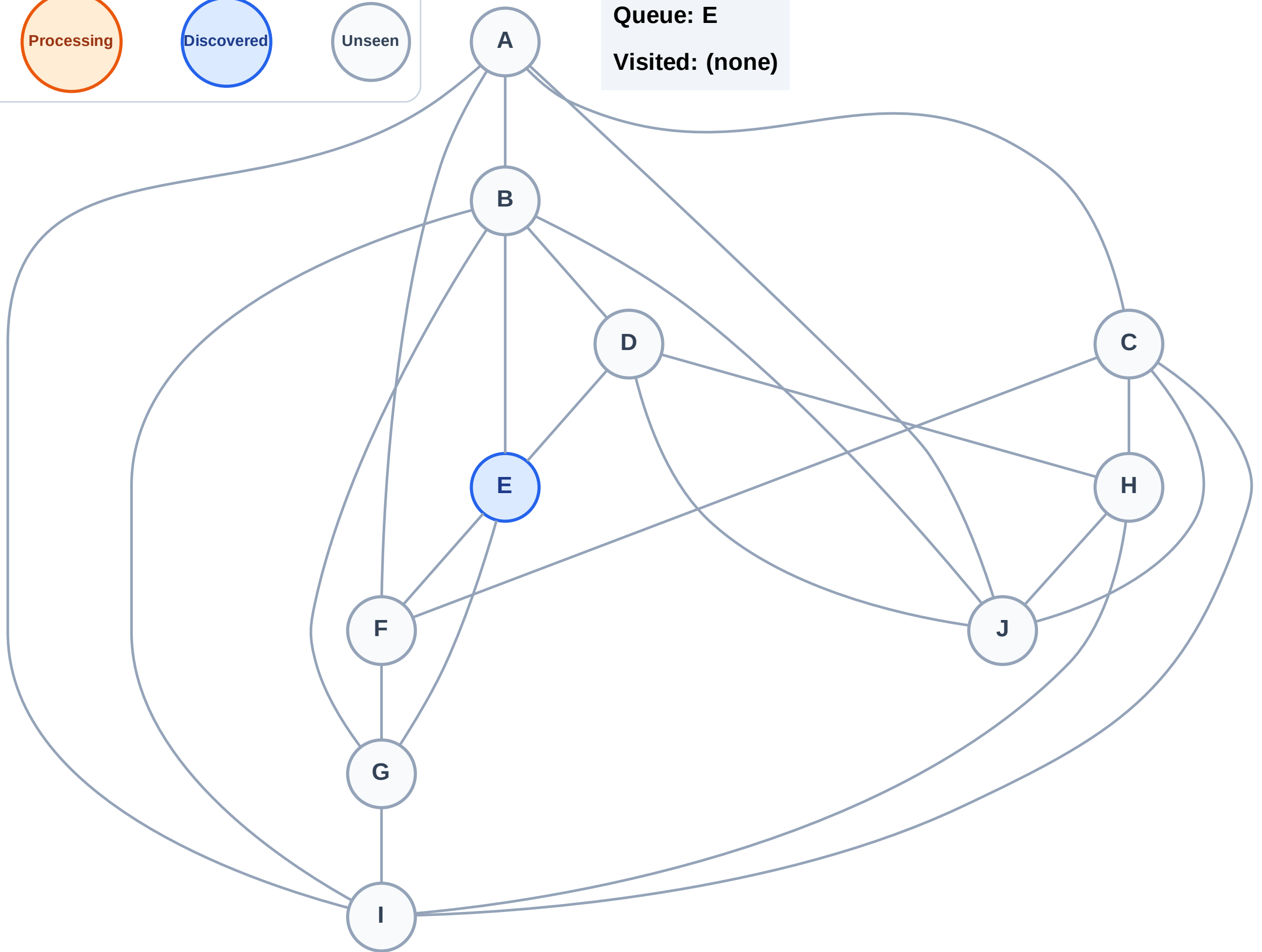
Step 1: Start at E

Legend



Queue: E

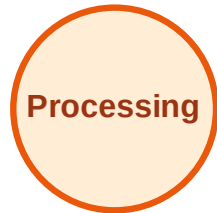
Visited: (none)



BFS Traversal

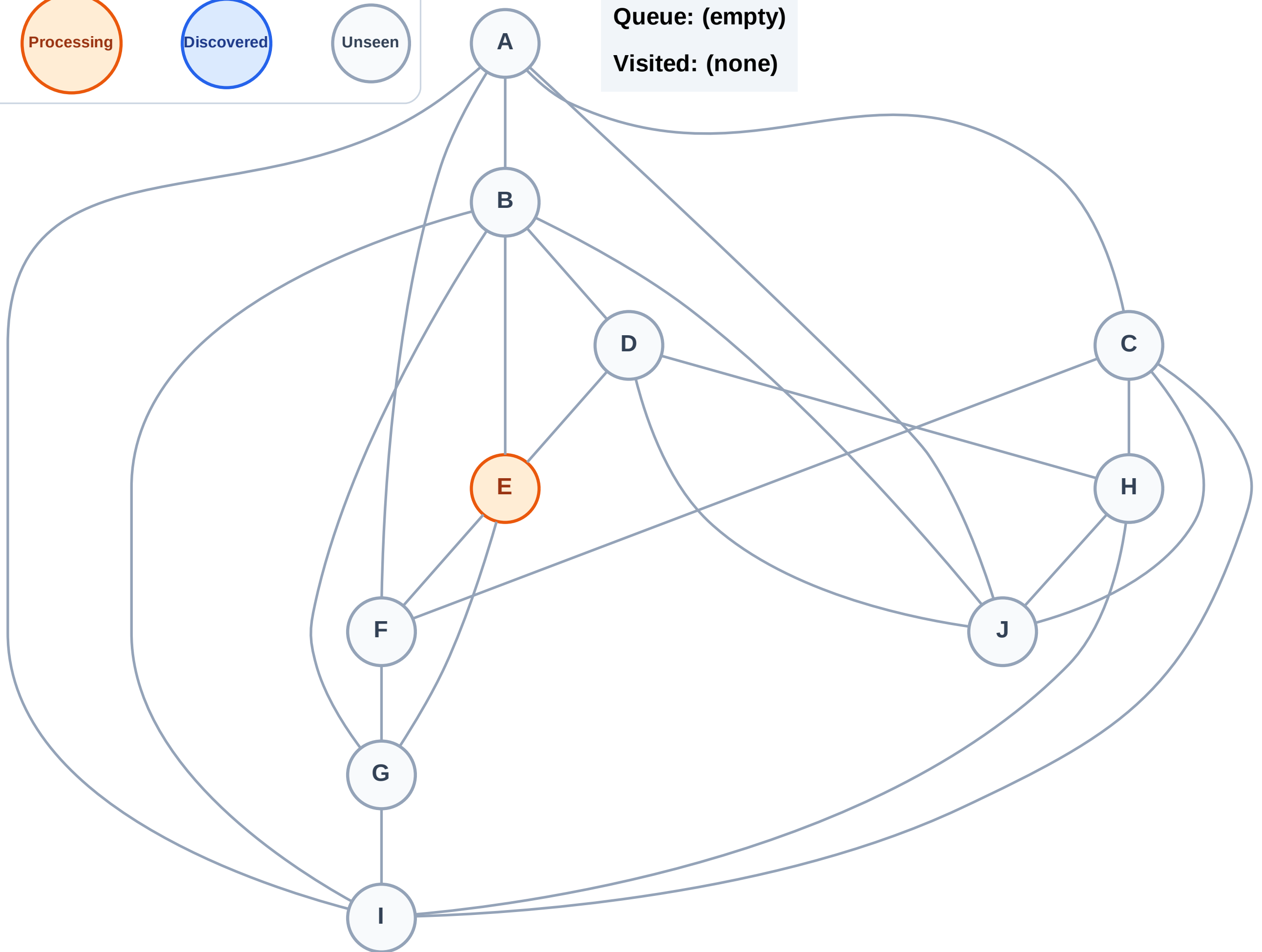
Step 2: Process node E

Legend



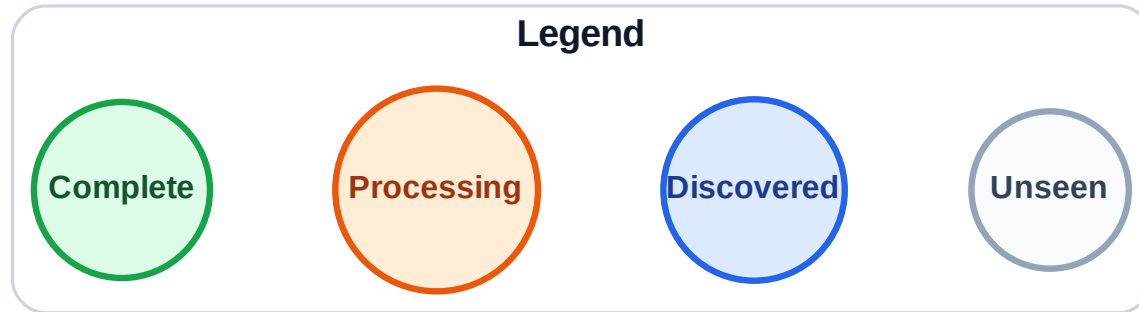
Queue: (empty)

Visited: (none)

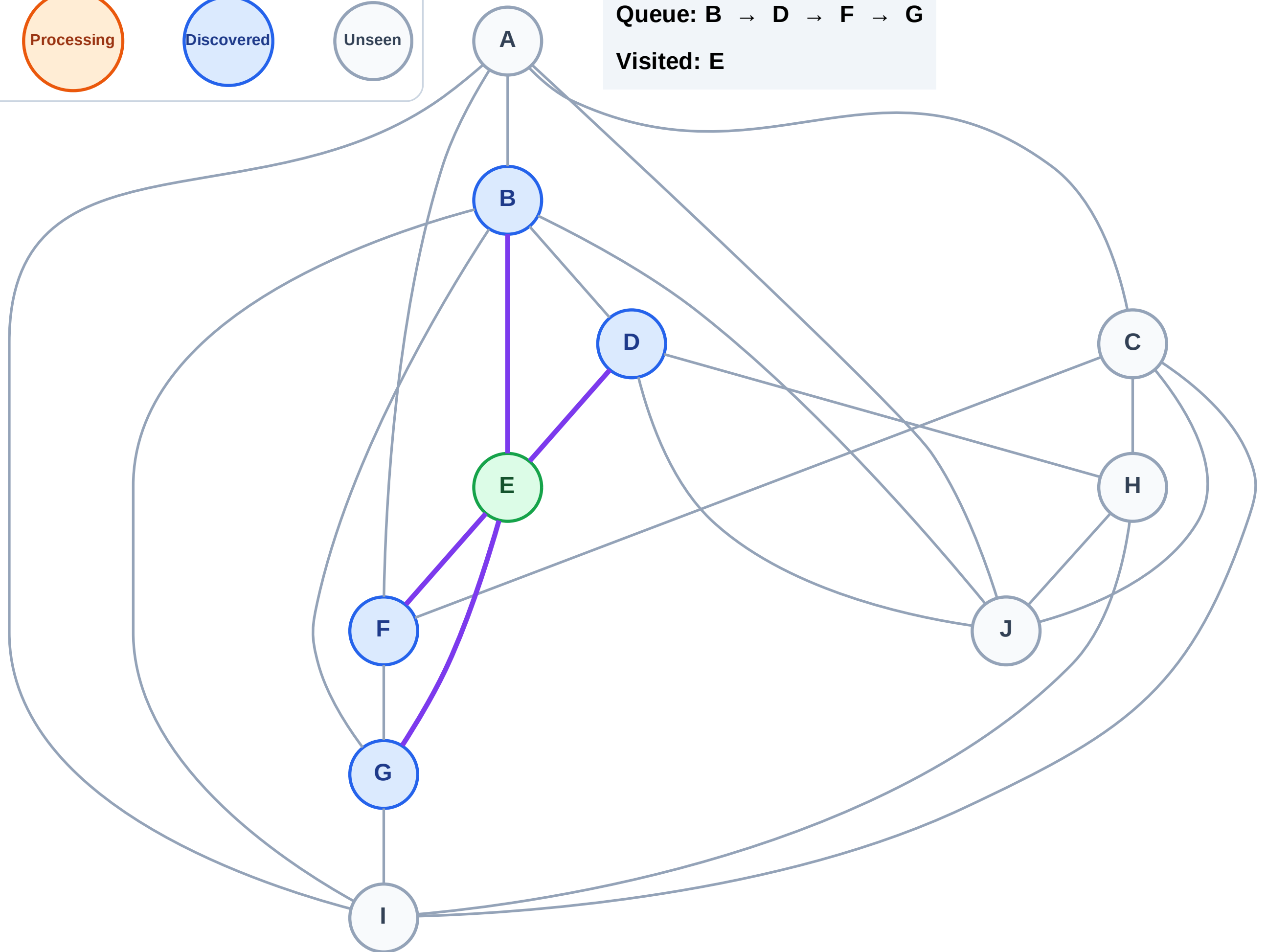


BFS Traversal

Step 3: Discover B, D, F, G from E



Queue: B → D → F → G
Visited: E



BFS Traversal

Step 4: Process node B

Legend

Complete

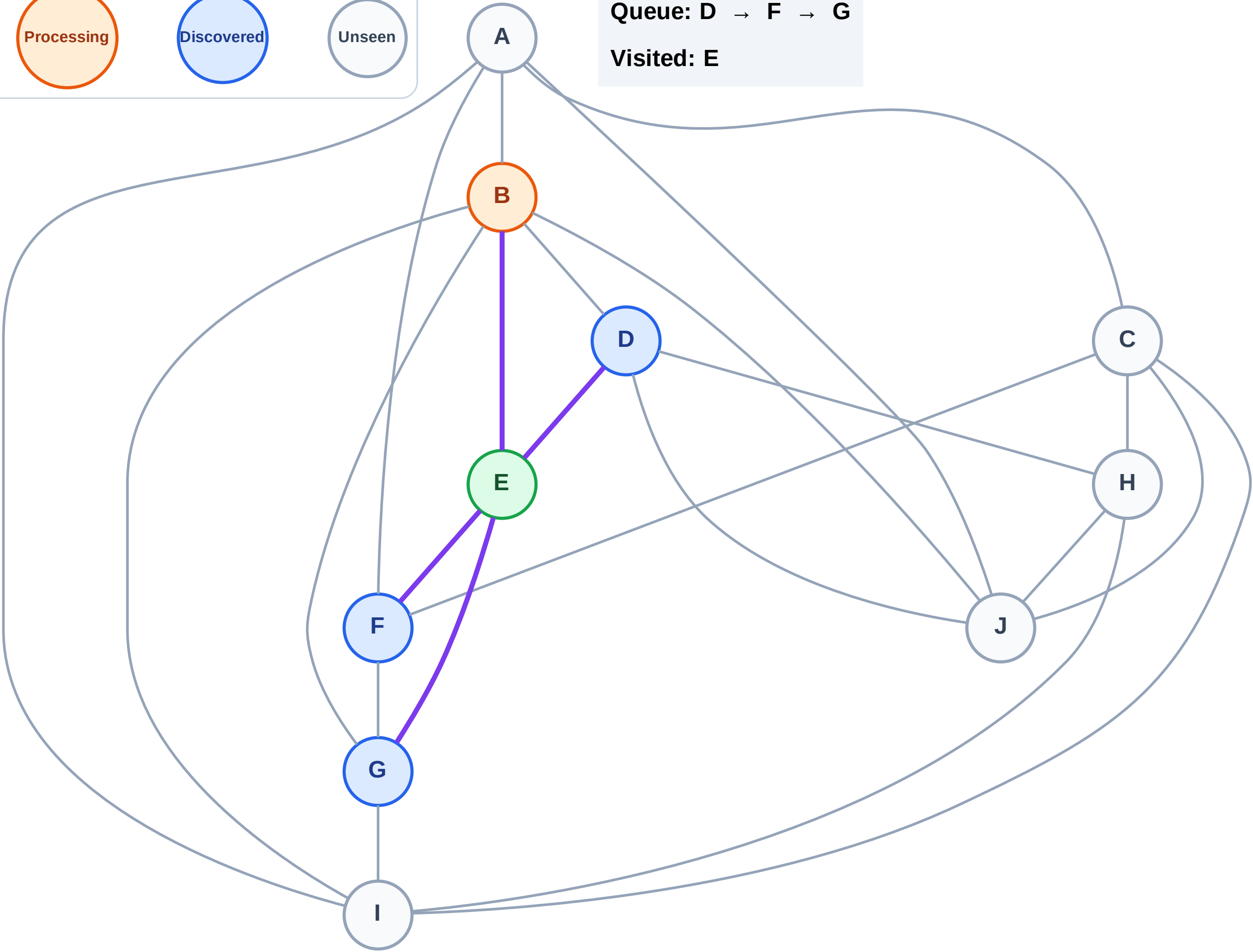
Processing

Discovered

Unseen

Queue: D → F → G

Visited: E



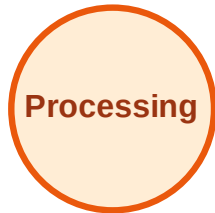
BFS Traversal

Step 5: Discover A, I, J from B

Legend



Complete



Processing



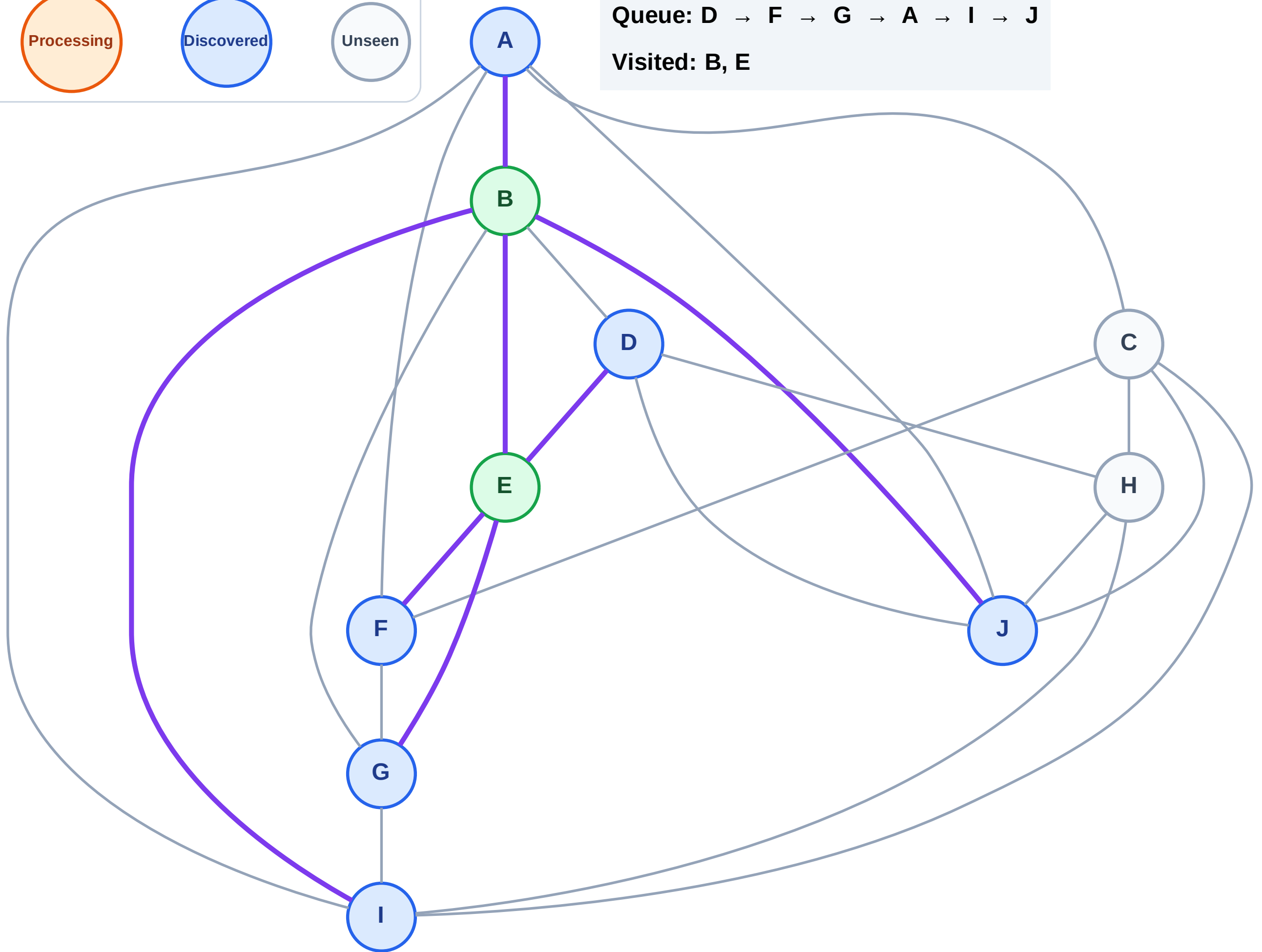
Discovered



Unseen

Queue: D → F → G → A → I → J

Visited: B, E



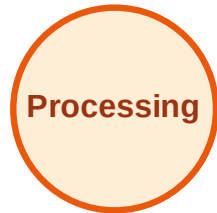
BFS Traversal

Step 6: Process node D

Legend



Complete



Processing



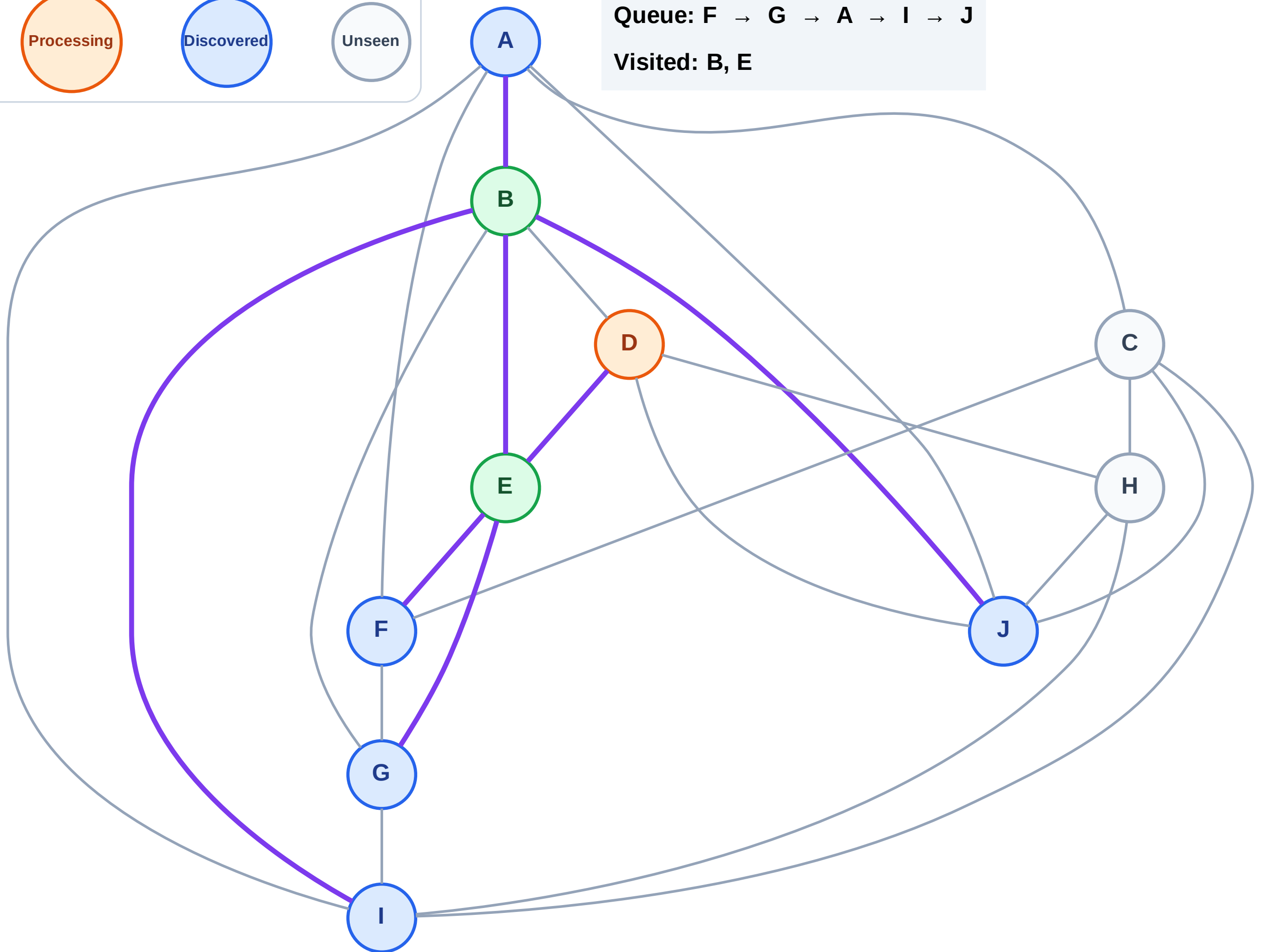
Discovered



Unseen

Queue: F → G → A → I → J

Visited: B, E



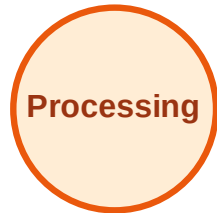
BFS Traversal

Step 7: Discover H from D

Legend



Complete



Processing



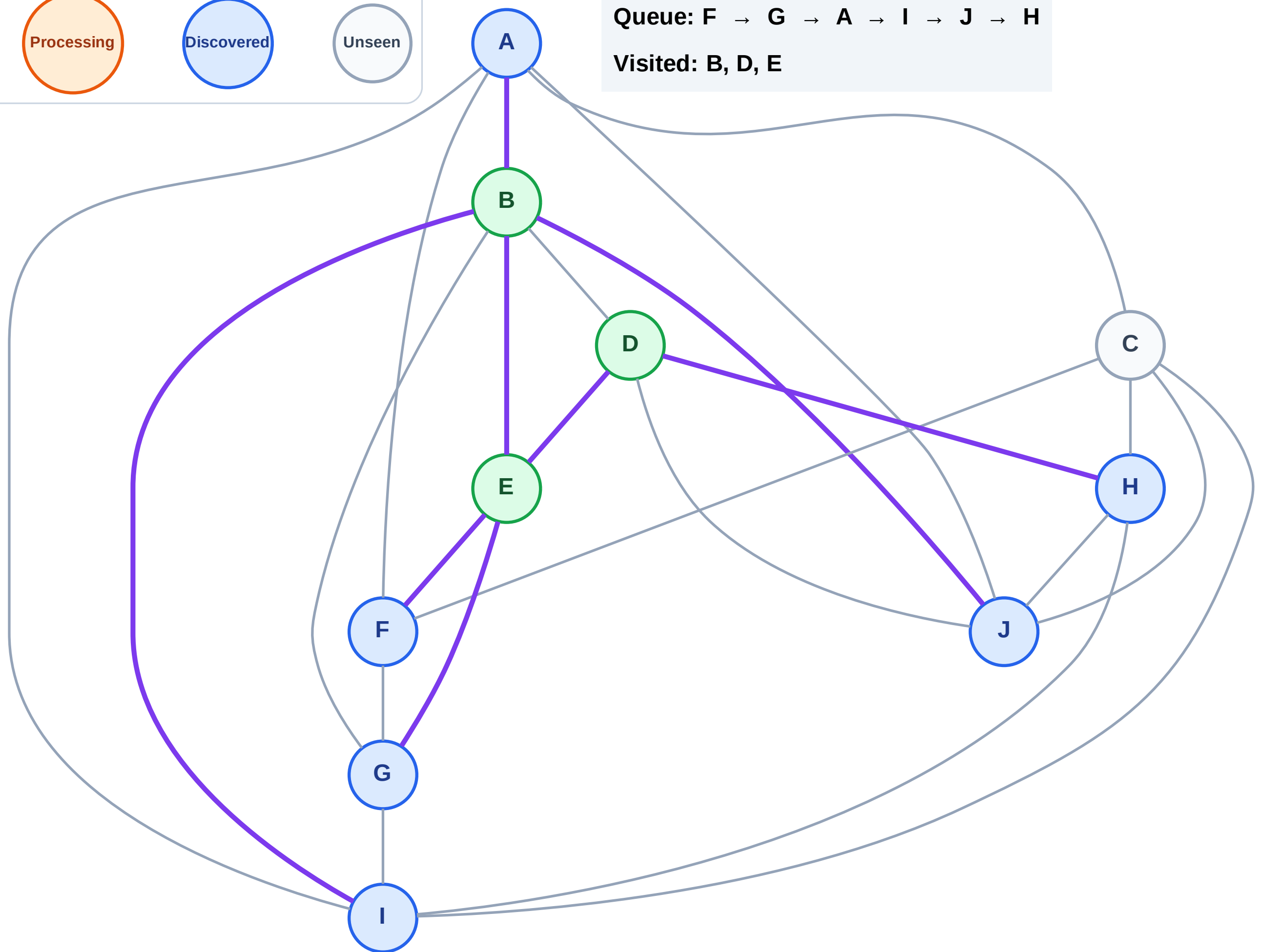
Discovered



Unseen

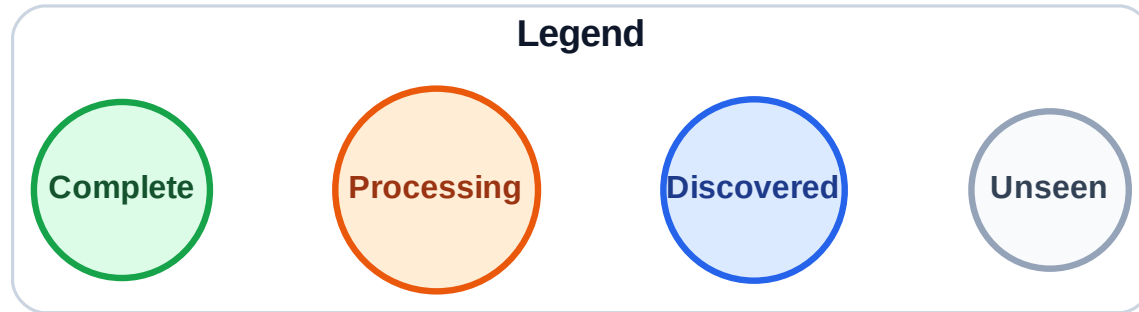
Queue: F → G → A → I → J → H

Visited: B, D, E



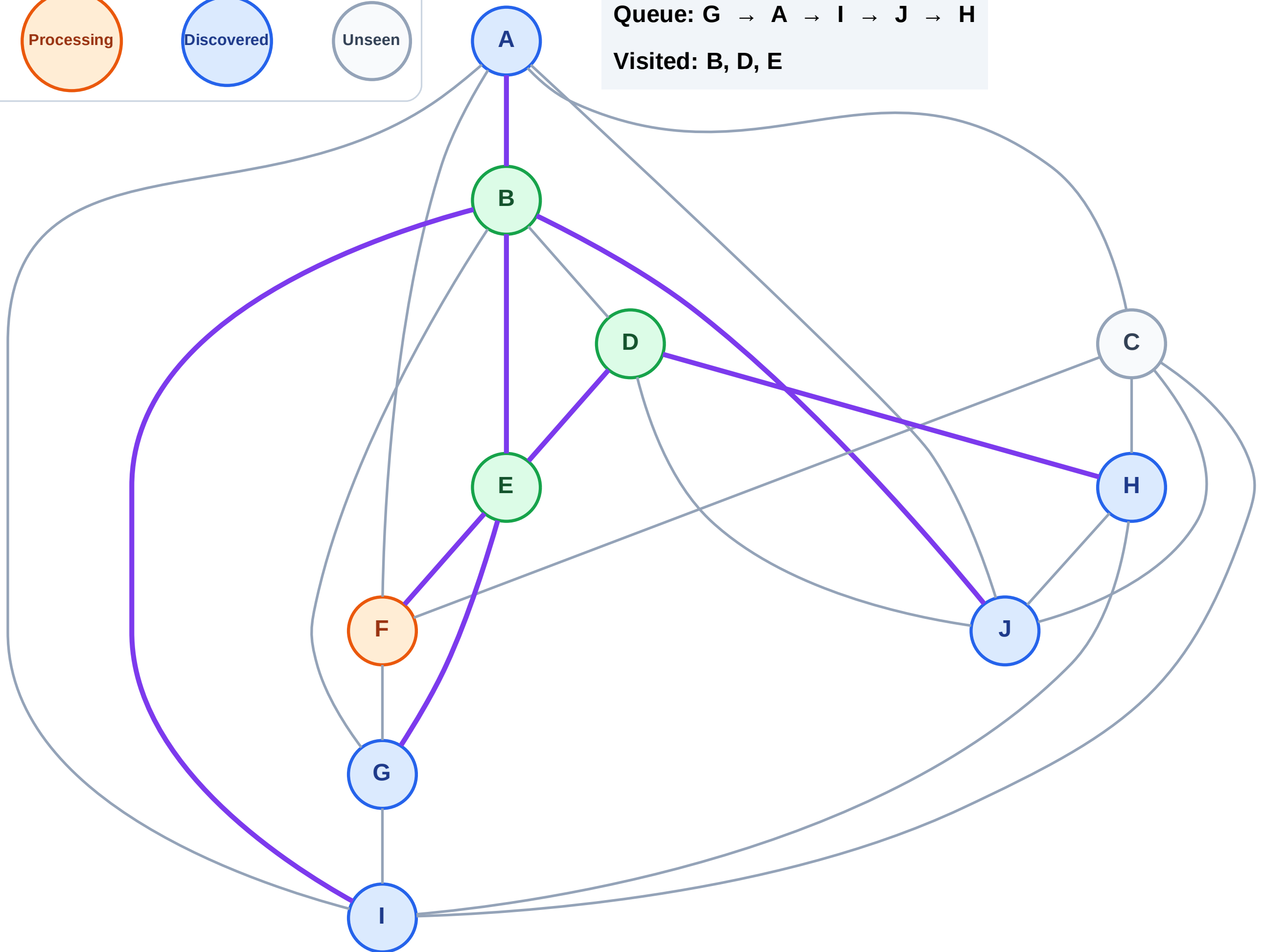
BFS Traversal

Step 8: Process node F



Queue: G → A → I → J → H

Visited: B, D, E



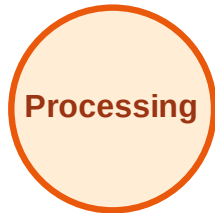
BFS Traversal

Step 9: Discover C from F

Legend



Complete



Processing



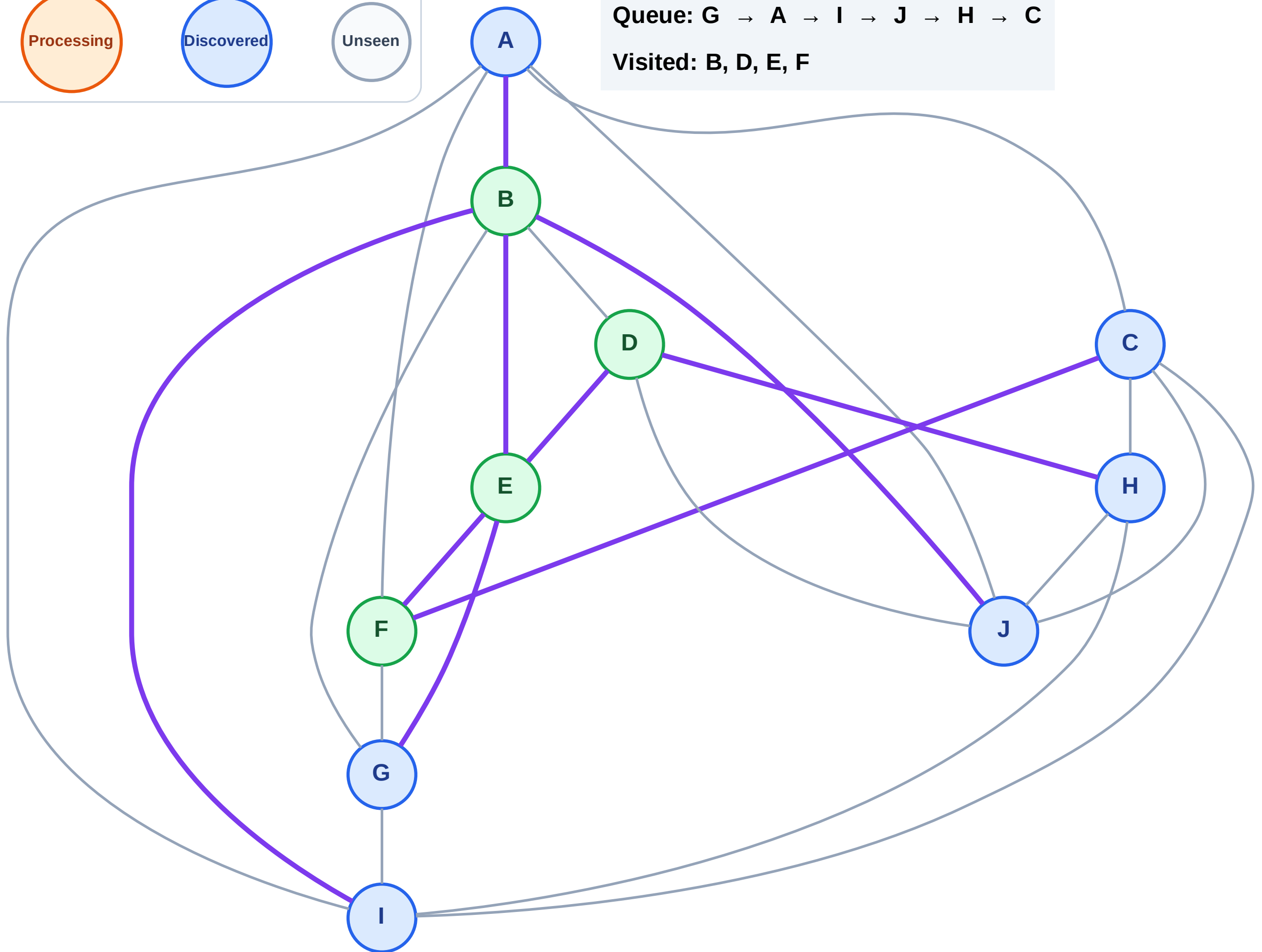
Discovered



Unseen

Queue: G → A → I → J → H → C

Visited: B, D, E, F



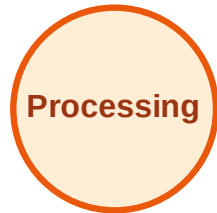
BFS Traversal

Step 10: Process node G

Legend



Complete



Processing



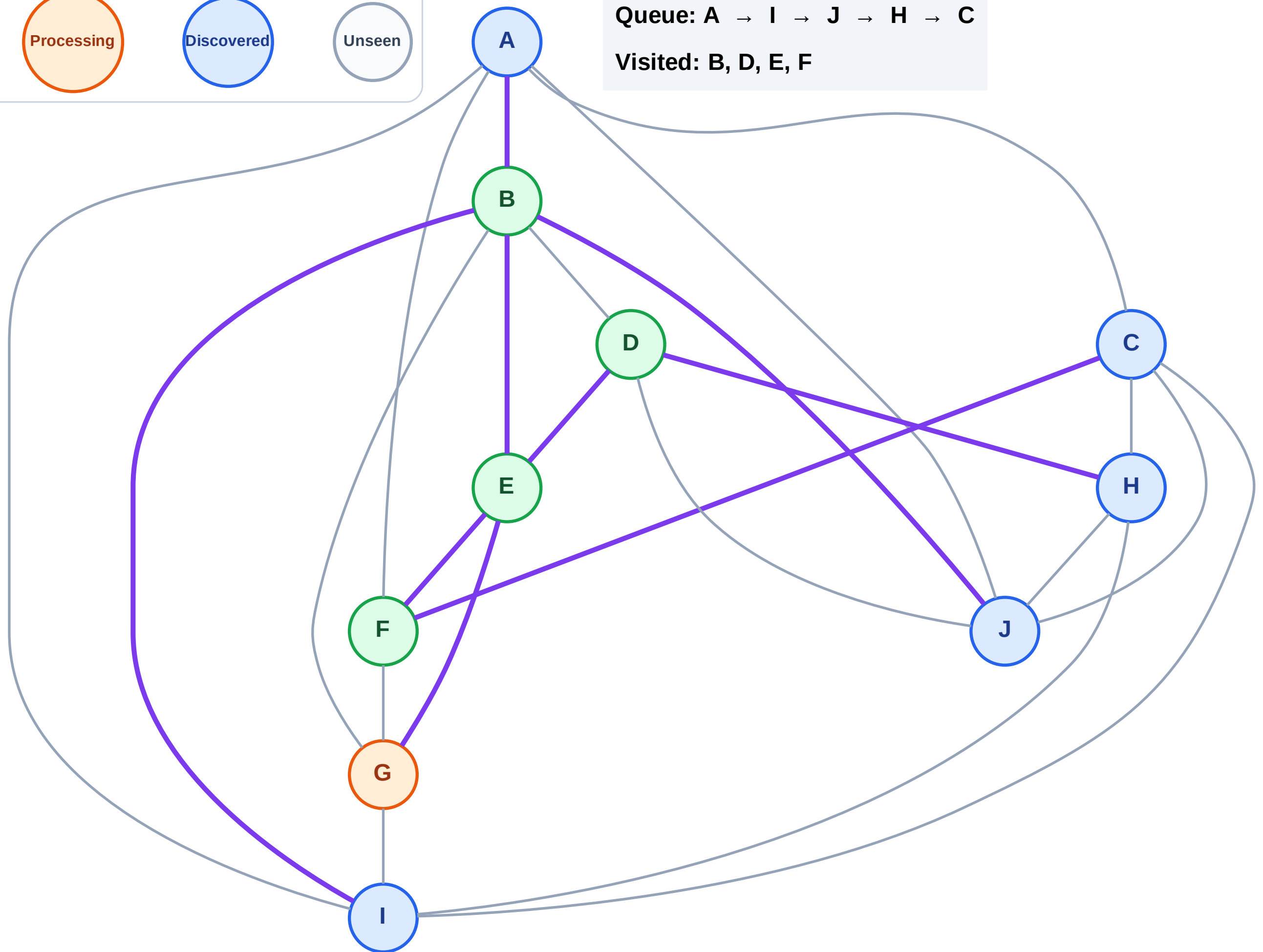
Discovered



Unseen

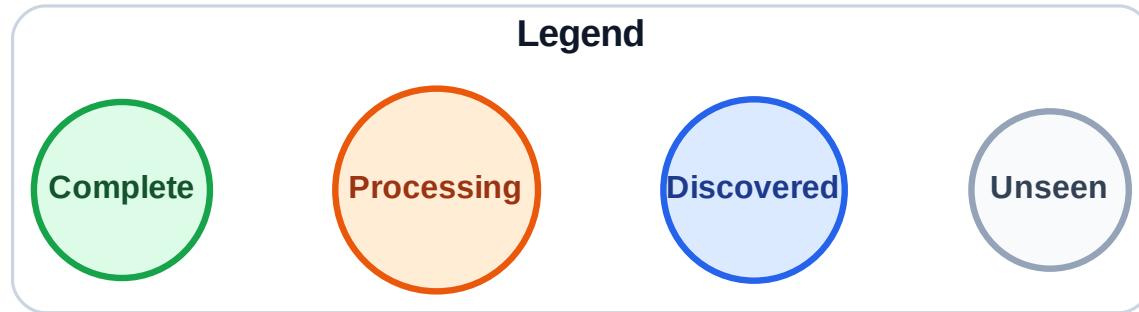
Queue: A → I → J → H → C

Visited: B, D, E, F

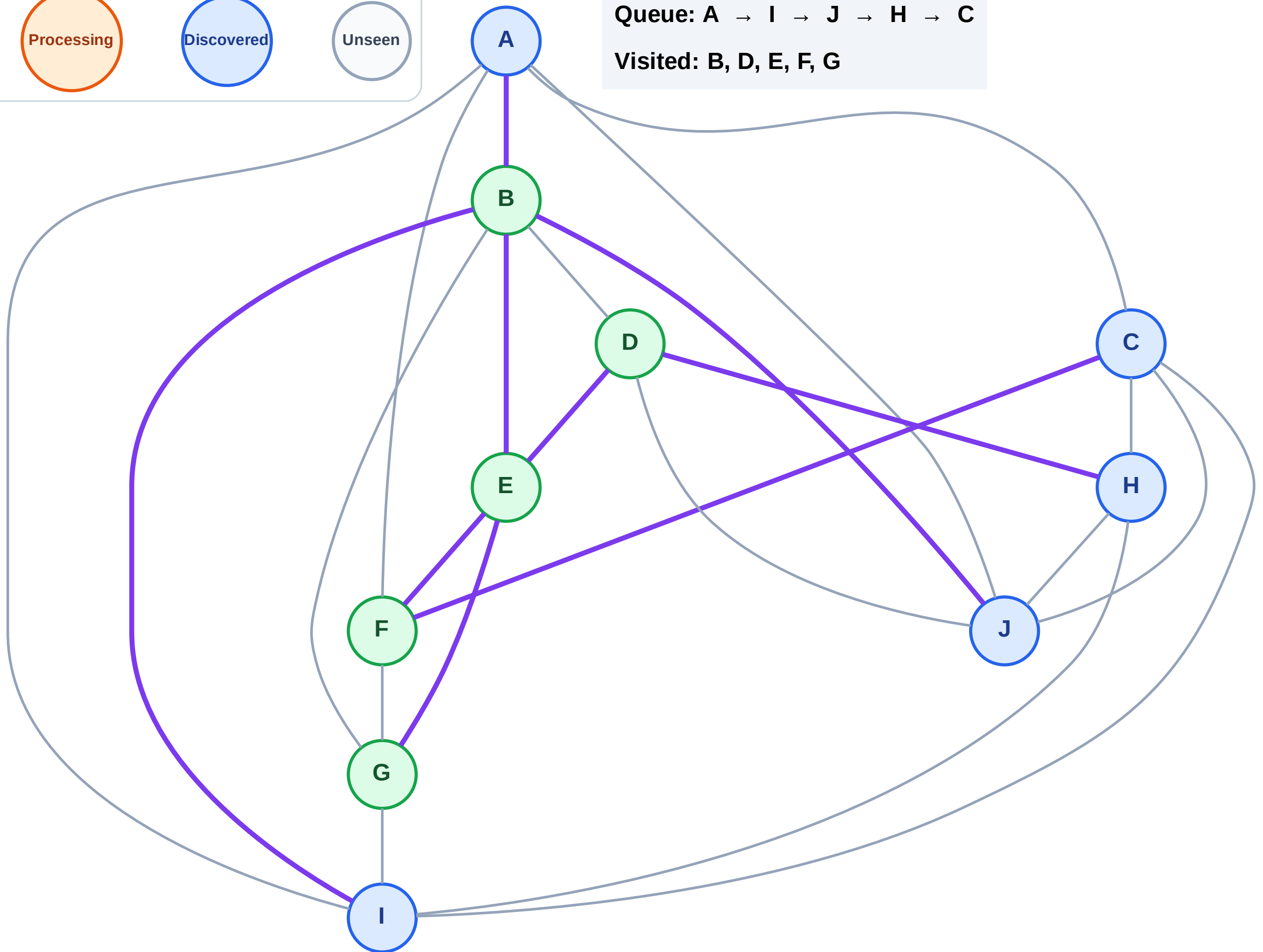


BFS Traversal

Step 11: No new neighbors from G



Queue: A → I → J → H → C
Visited: B, D, E, F, G



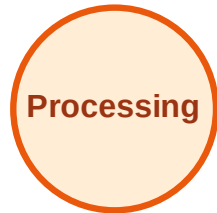
BFS Traversal

Step 12: Process node A

Legend



Complete



Processing



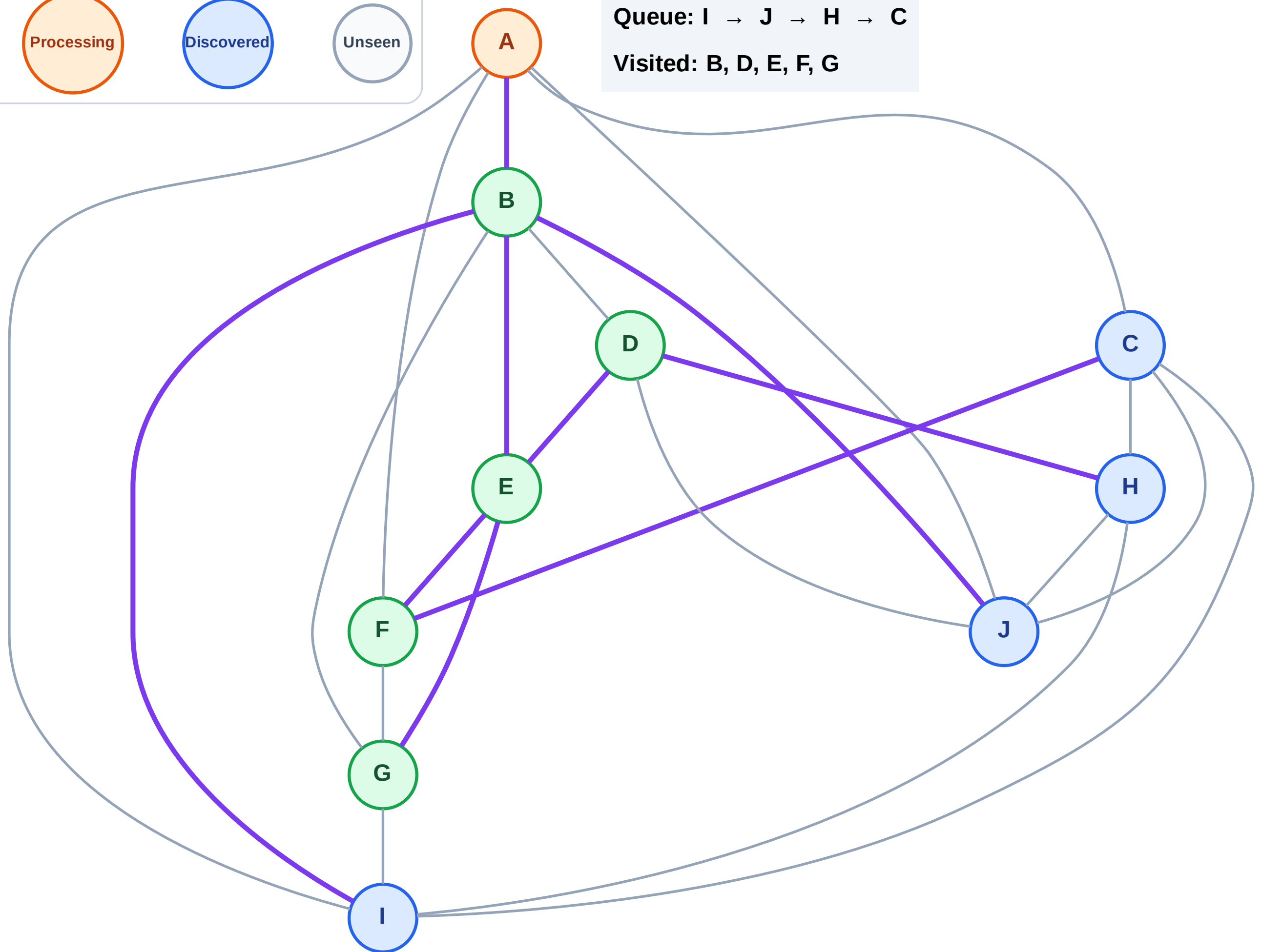
Discovered



Unseen

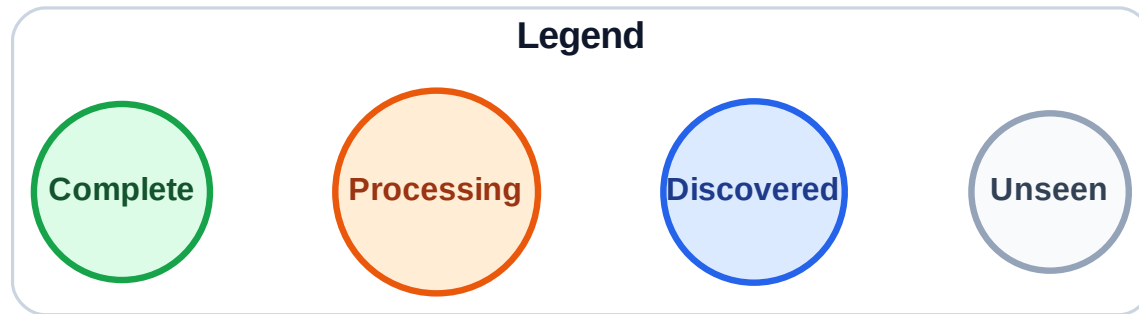
Queue: I → J → H → C

Visited: B, D, E, F, G



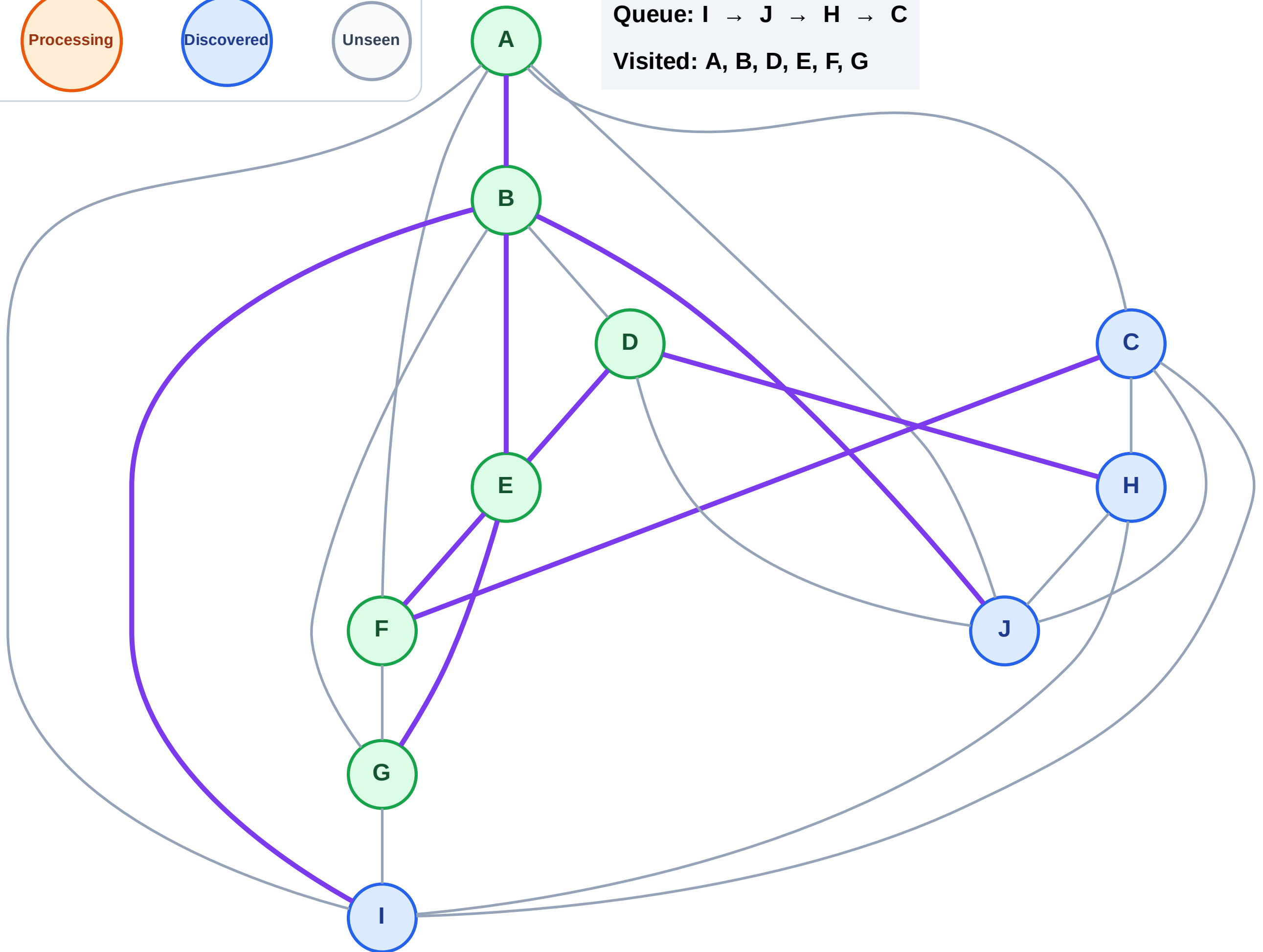
BFS Traversal

Step 13: No new neighbors from A



Queue: I → J → H → C

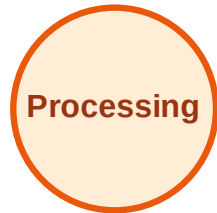
Visited: A, B, D, E, F, G



BFS Traversal

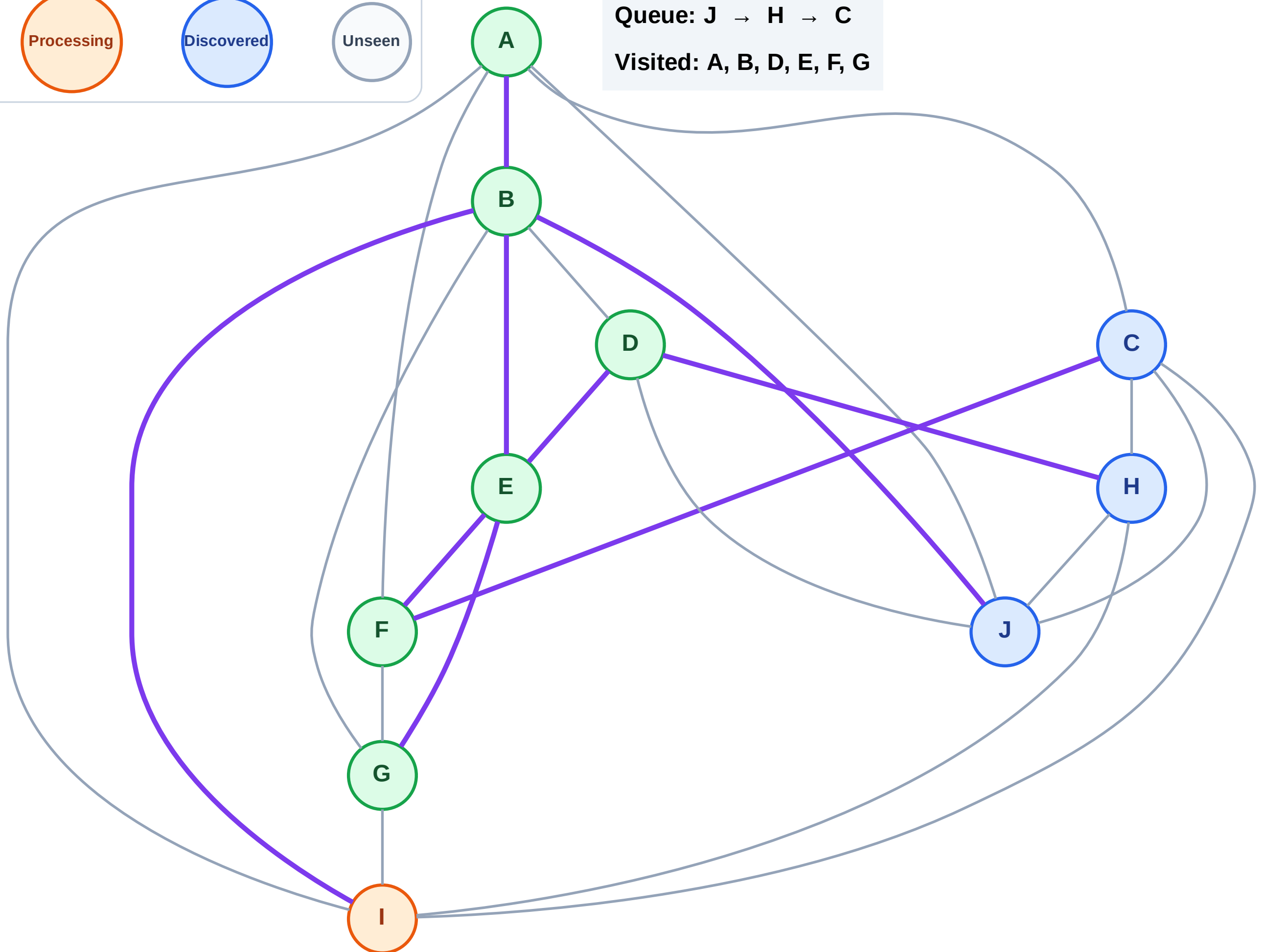
Step 14: Process node I

Legend



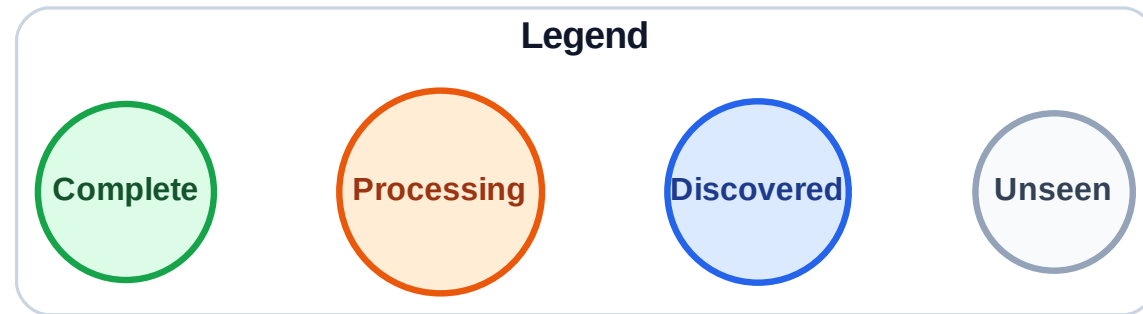
Queue: J → H → C

Visited: A, B, D, E, F, G

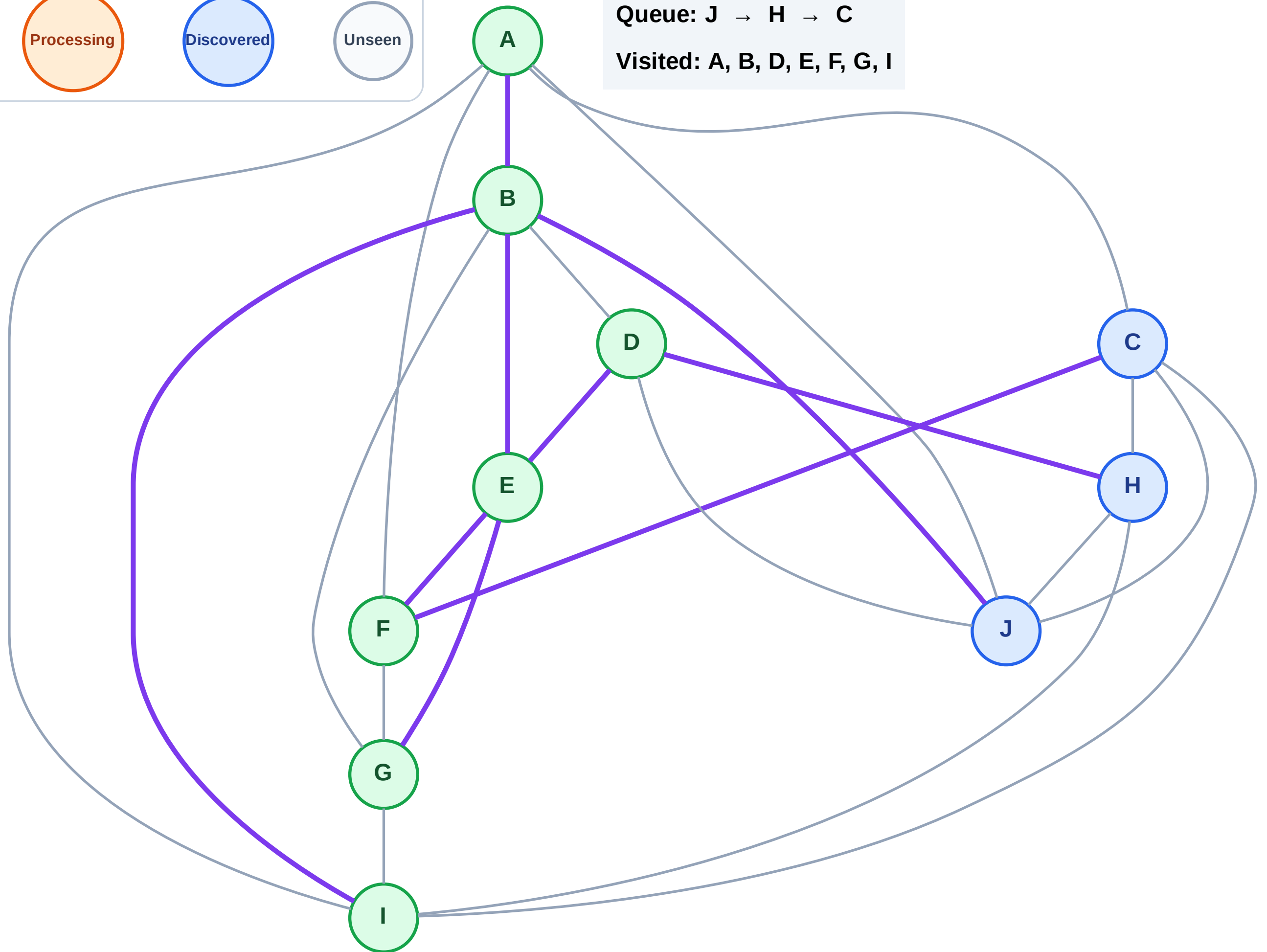


BFS Traversal

Step 15: No new neighbors from I



Queue: J → H → C
Visited: A, B, D, E, F, G, I



BFS Traversal

Step 16: Process node J

Legend

Complete

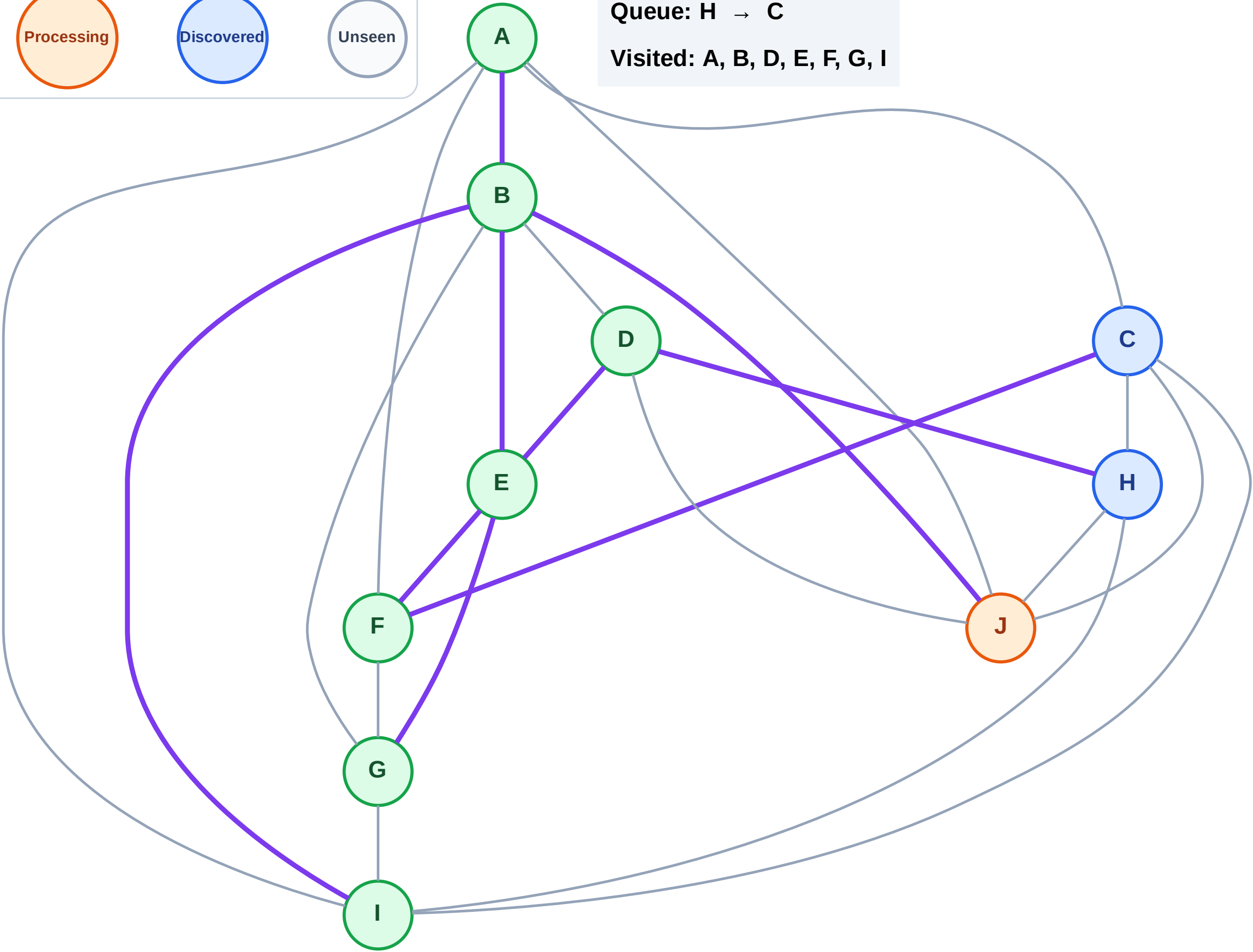
Processing

Discovered

Unseen

Queue: H → C

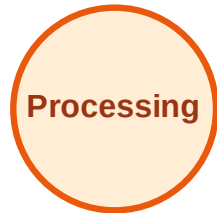
Visited: A, B, D, E, F, G, I



BFS Traversal

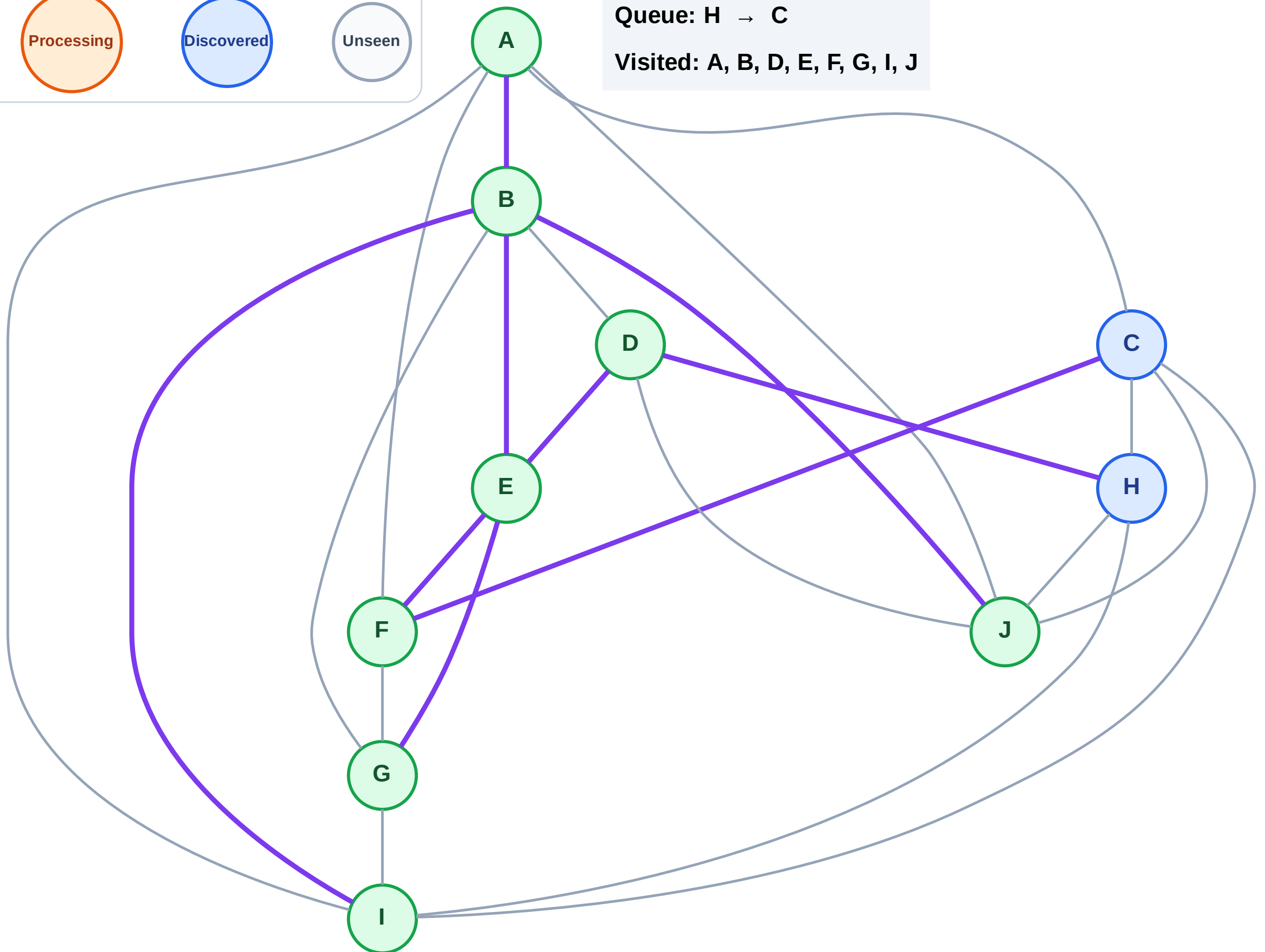
Step 17: No new neighbors from J

Legend



Queue: H → C

Visited: A, B, D, E, F, G, I, J



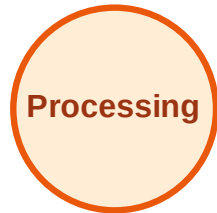
BFS Traversal

Step 18: Process node H

Legend



Complete



Processing



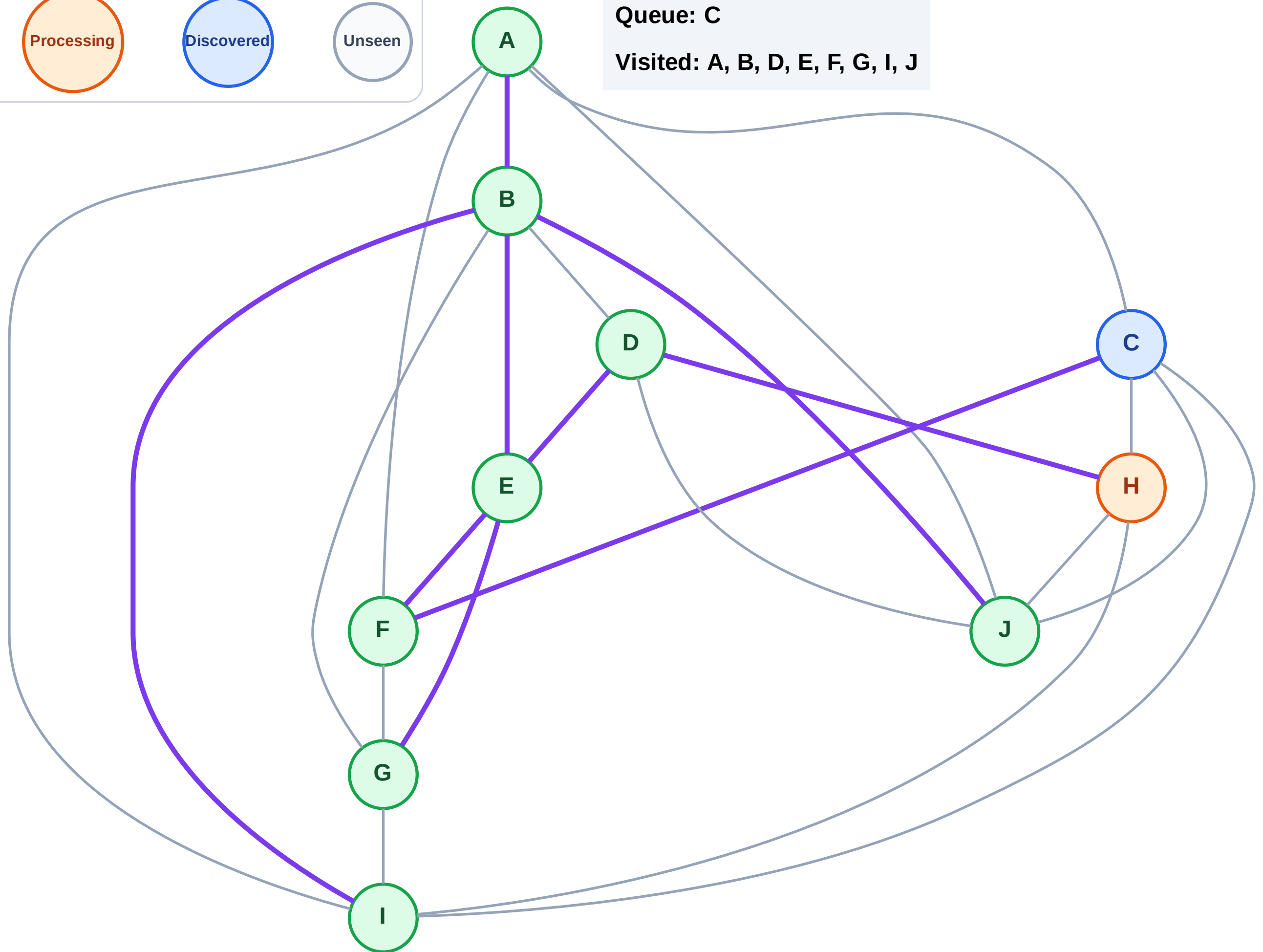
Discovered



Unseen

Queue: C

Visited: A, B, D, E, F, G, I, J



BFS Traversal

Step 19: No new neighbors from H

Legend

Complete

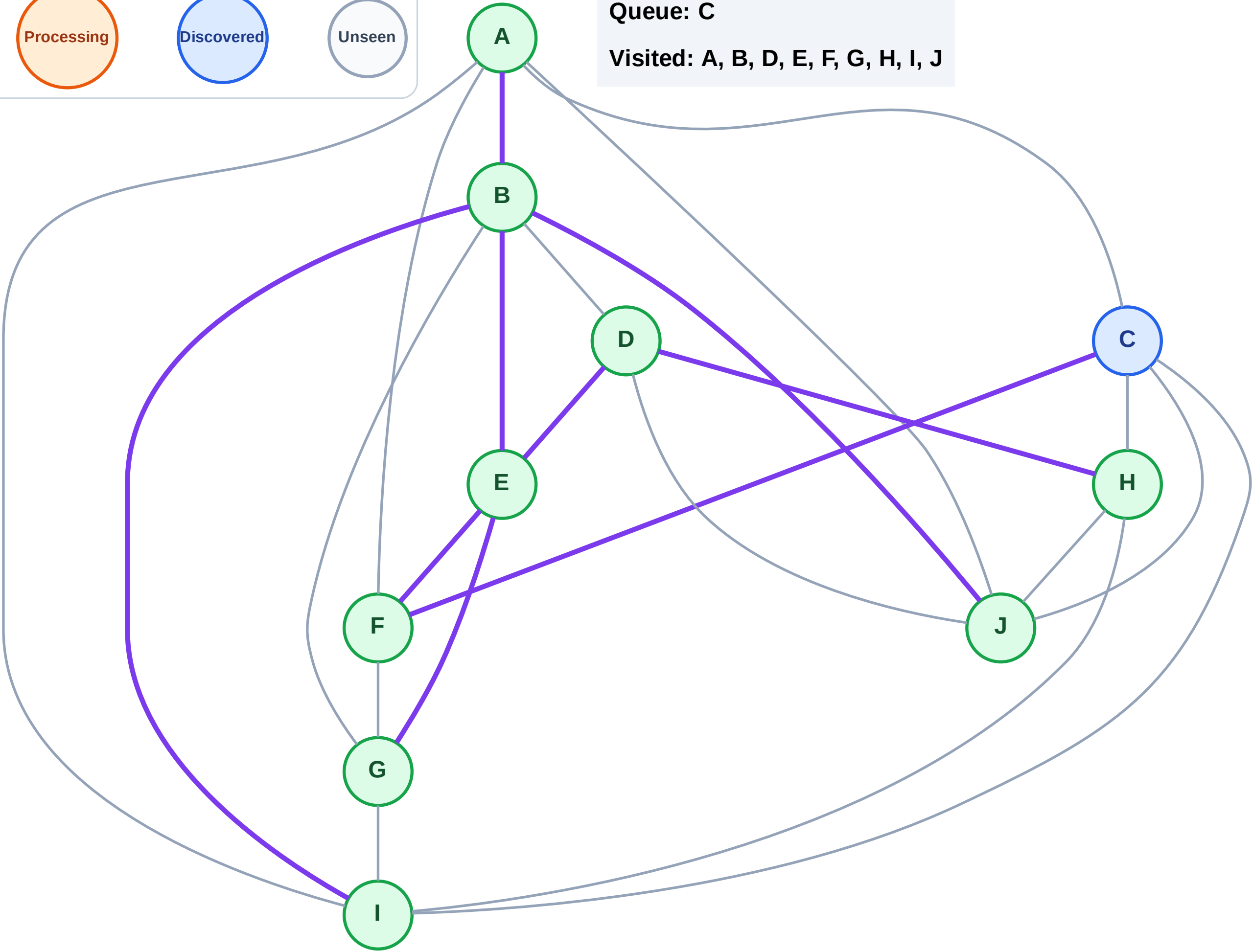
Processing

Discovered

Unseen

Queue: C

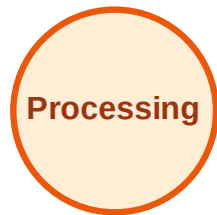
Visited: A, B, D, E, F, G, H, I, J



BFS Traversal

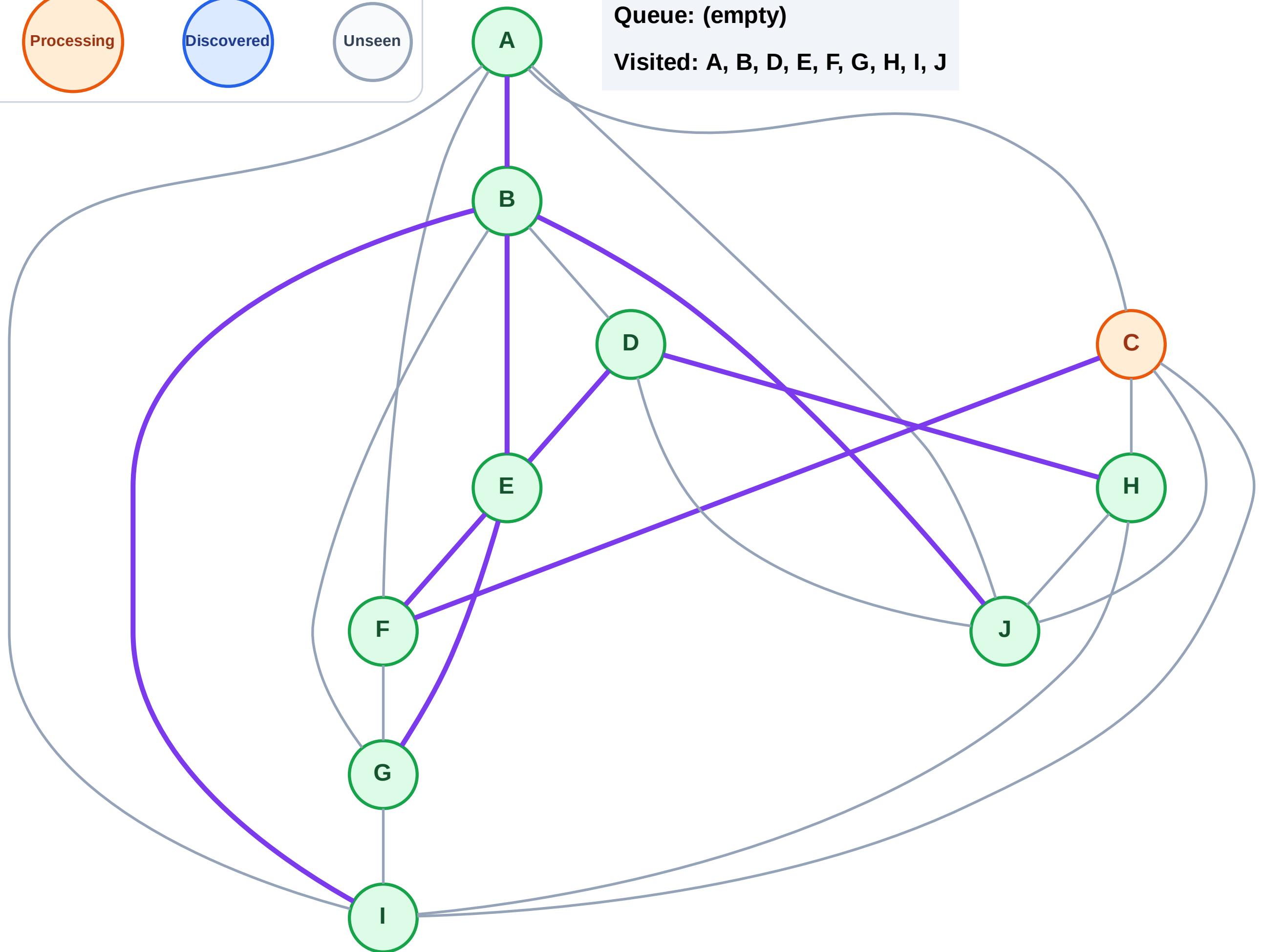
Step 20: Process node C

Legend



Queue: (empty)

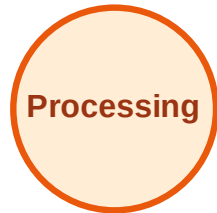
Visited: A, B, D, E, F, G, H, I, J



BFS Traversal

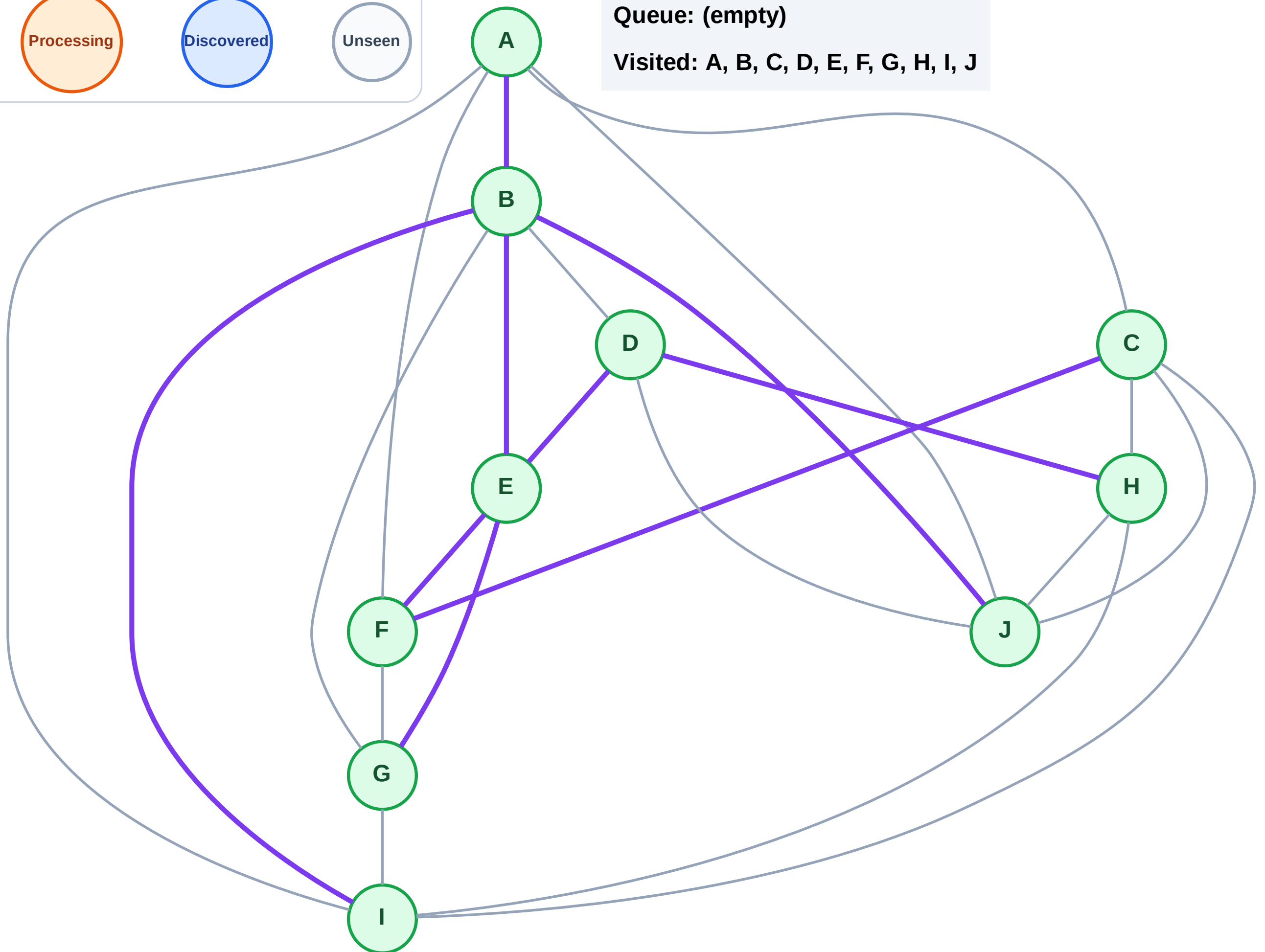
Step 21: No new neighbors from C

Legend



Queue: (empty)

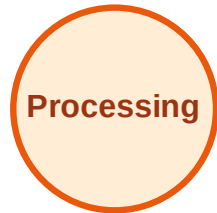
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

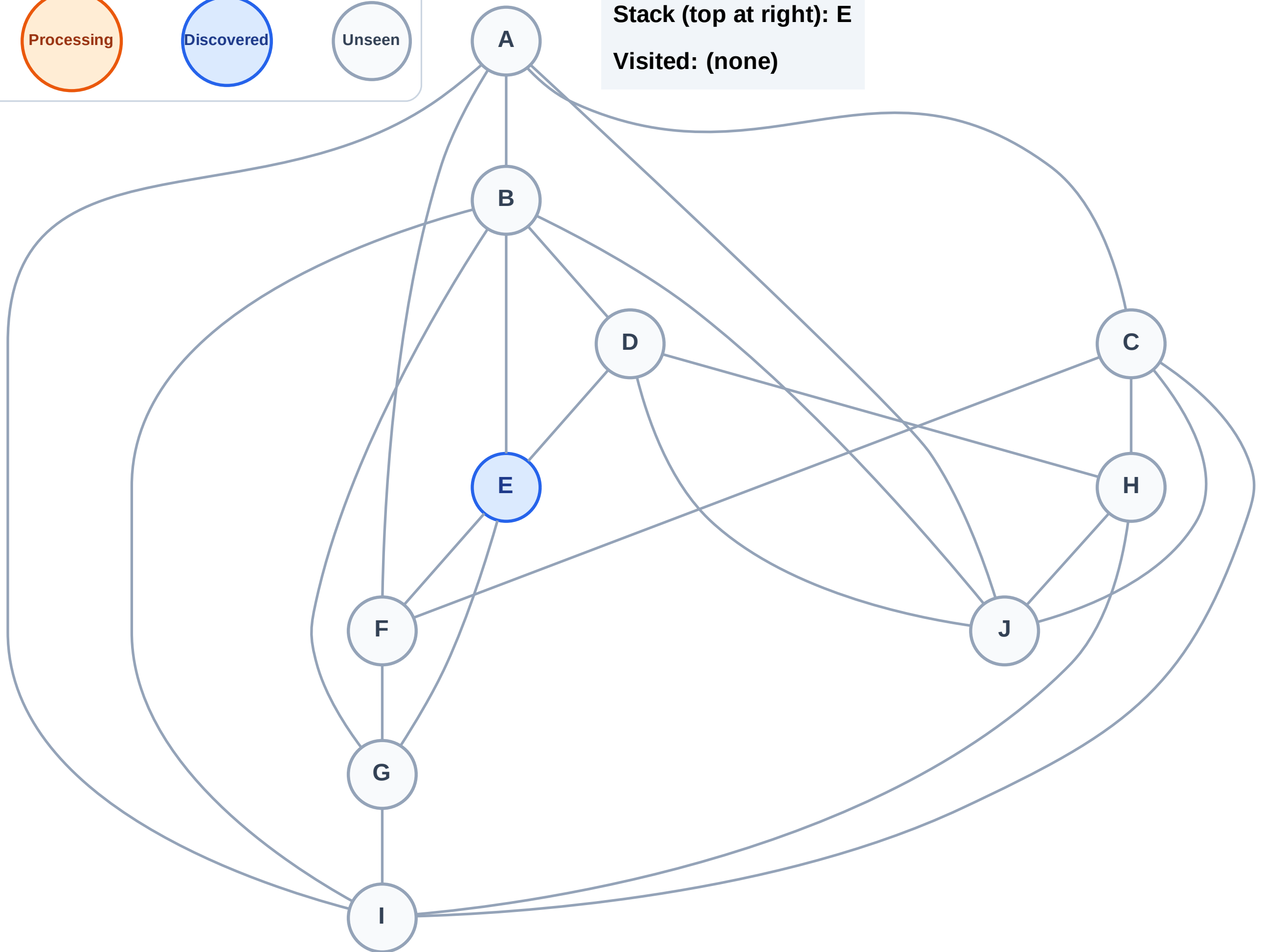
Step 1: Start at E

Legend



Stack (top at right): E

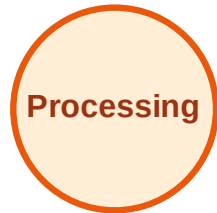
Visited: (none)



DFS Traversal

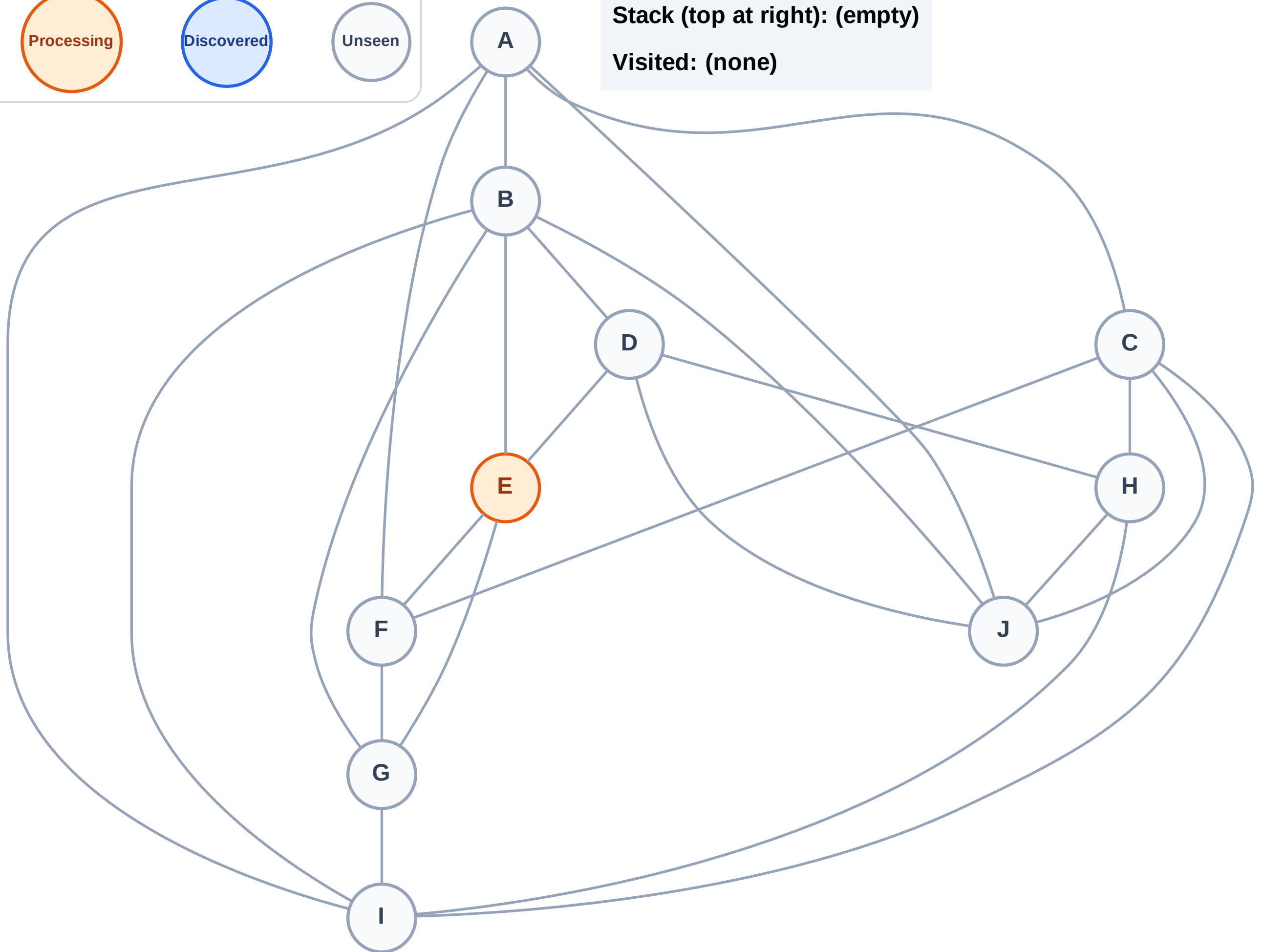
Step 2: Process node E

Legend



Stack (top at right): (empty)

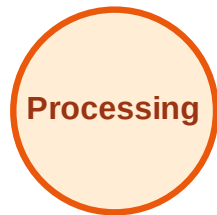
Visited: (none)



DFS Traversal

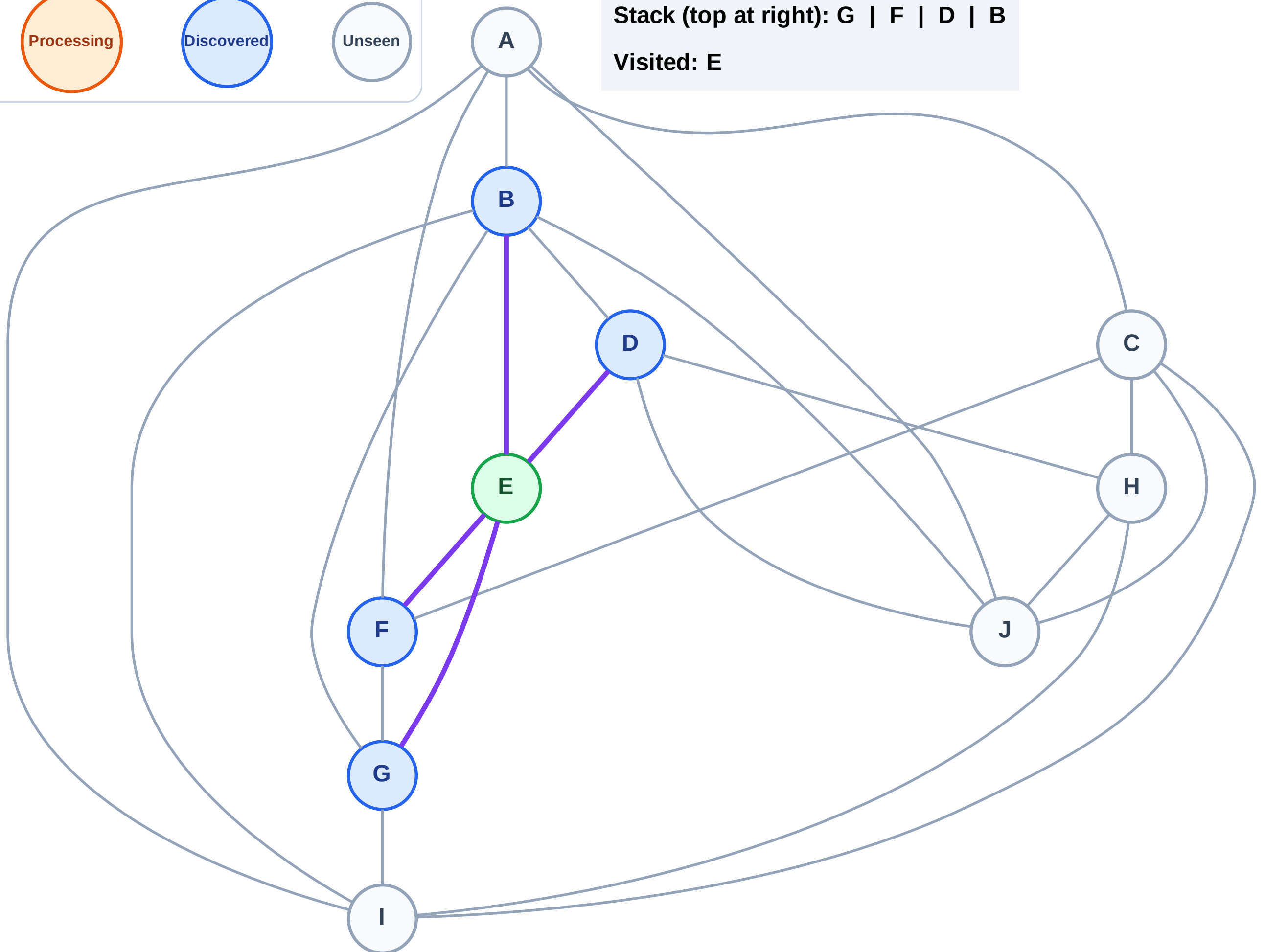
Step 3: Discover G, F, D, B from E

Legend



Stack (top at right): G | F | D | B

Visited: E



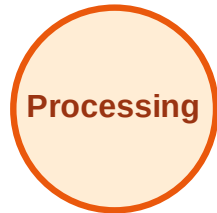
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



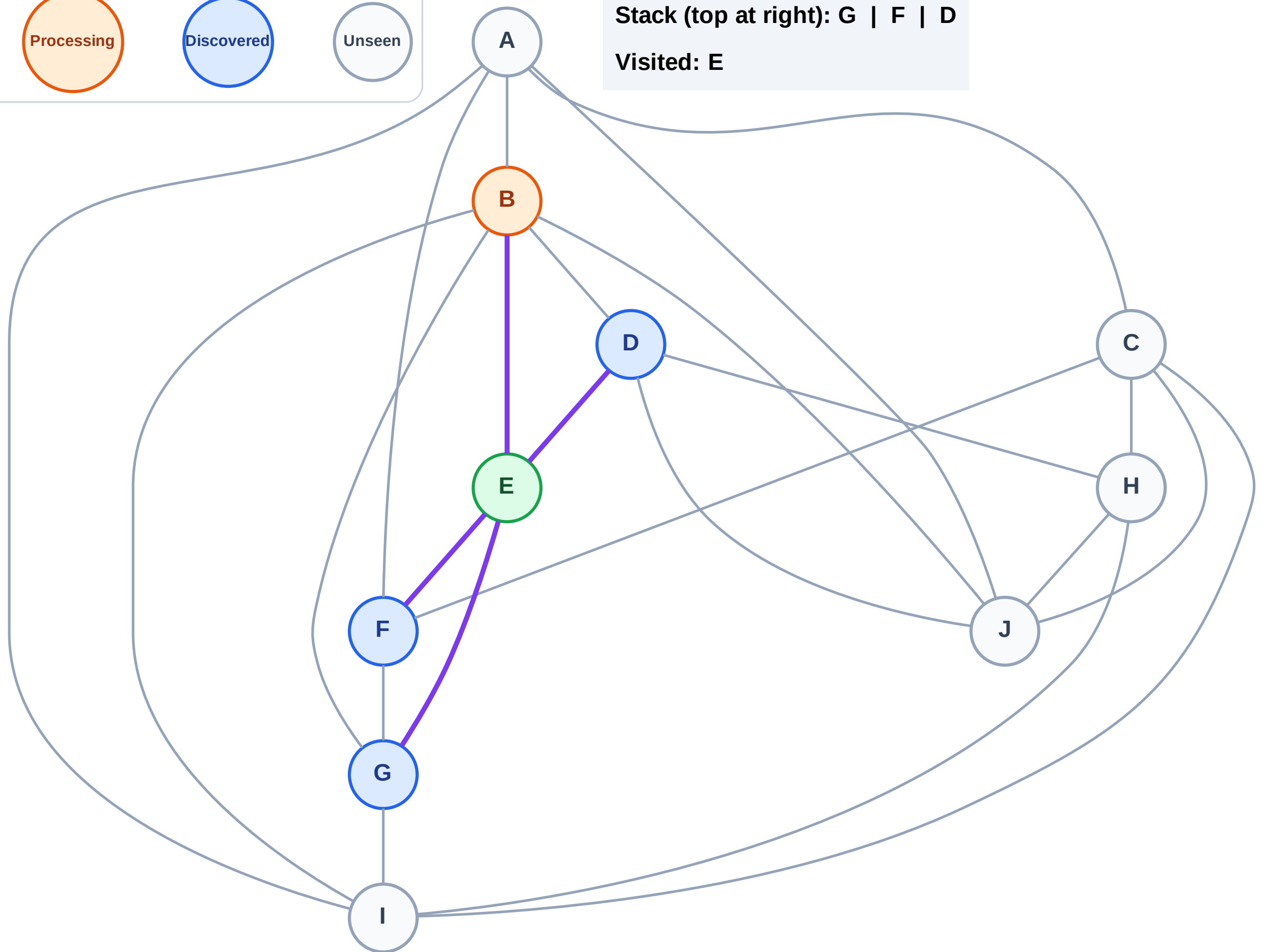
Discovered



Unseen

Stack (top at right): G | F | D

Visited: E



Step 5: Discover J, I, A from B

Stack (top at right): G | F | D | J | I | A

Legend

Processing

Unseen



DFS Traversal

Step 6: Process node A

Legend

Complete

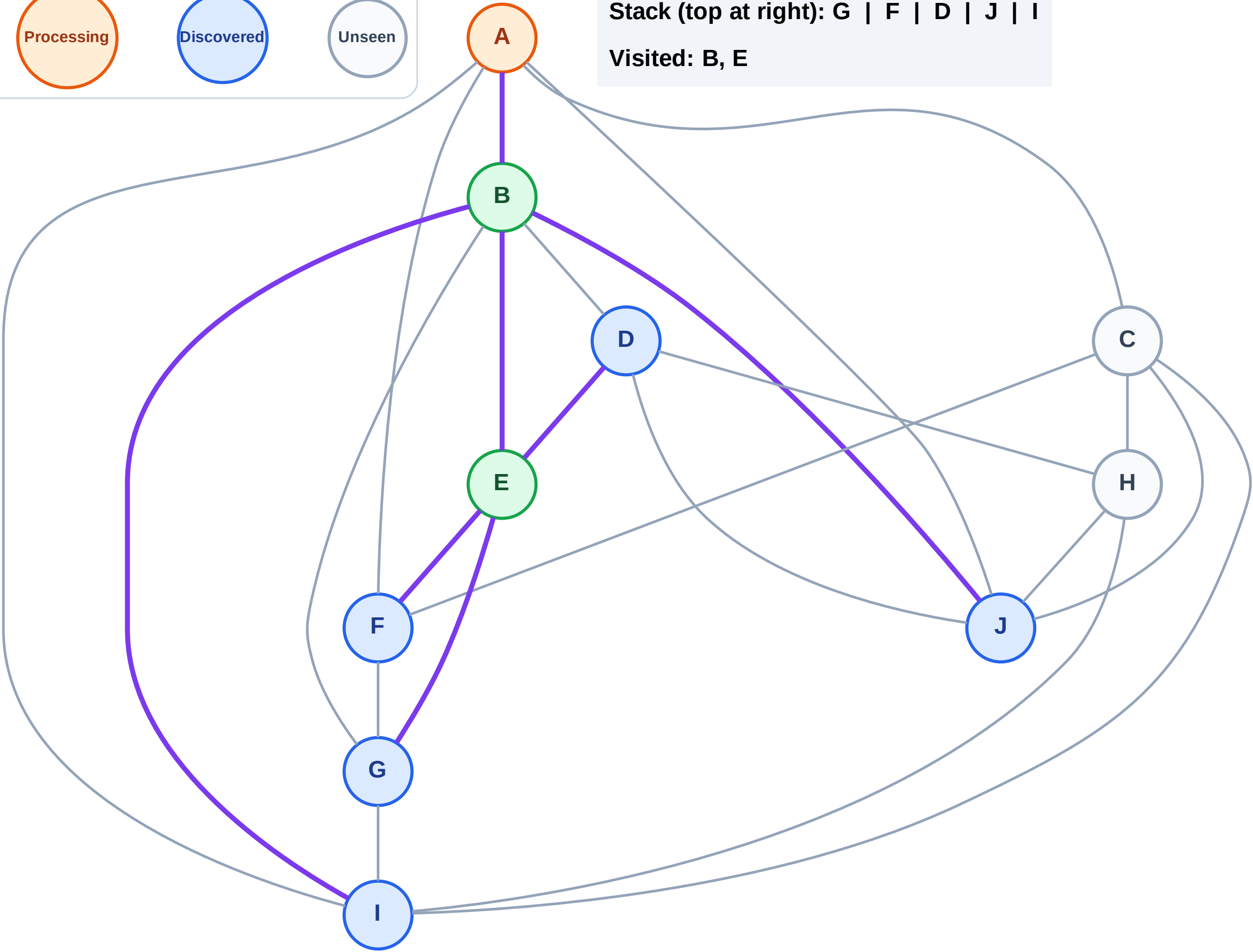
Processing

Discovered

Unseen

Stack (top at right): G | F | D | J | I

Visited: B, E



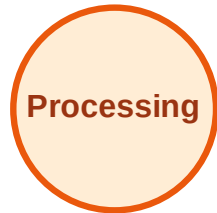
DFS Traversal

Step 7: Discover C from A

Legend



Complete



Processing



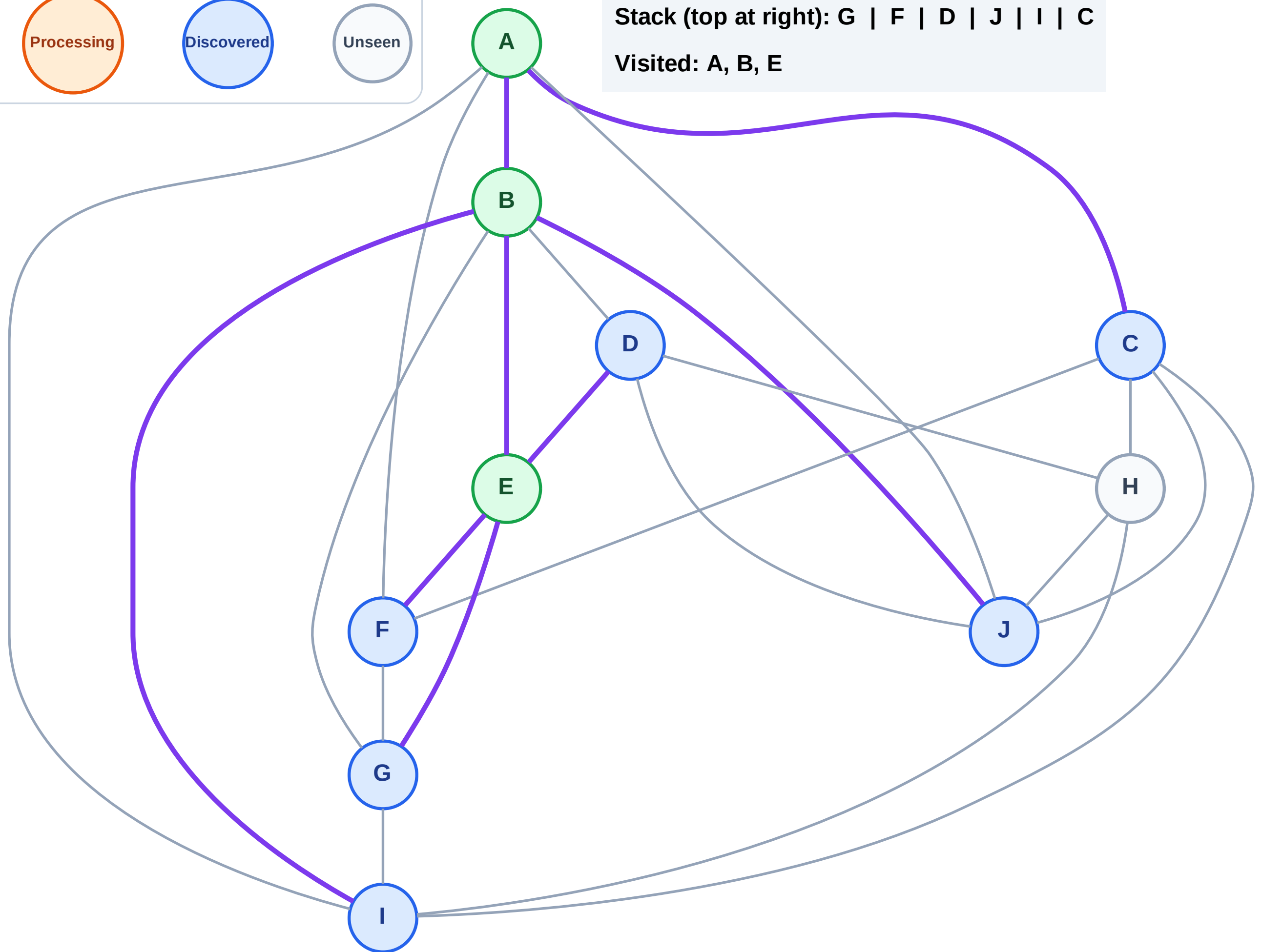
Discovered



Unseen

Stack (top at right): G | F | D | J | I | C

Visited: A, B, E



DFS Traversal

Step 8: Process node C

Legend

Complete

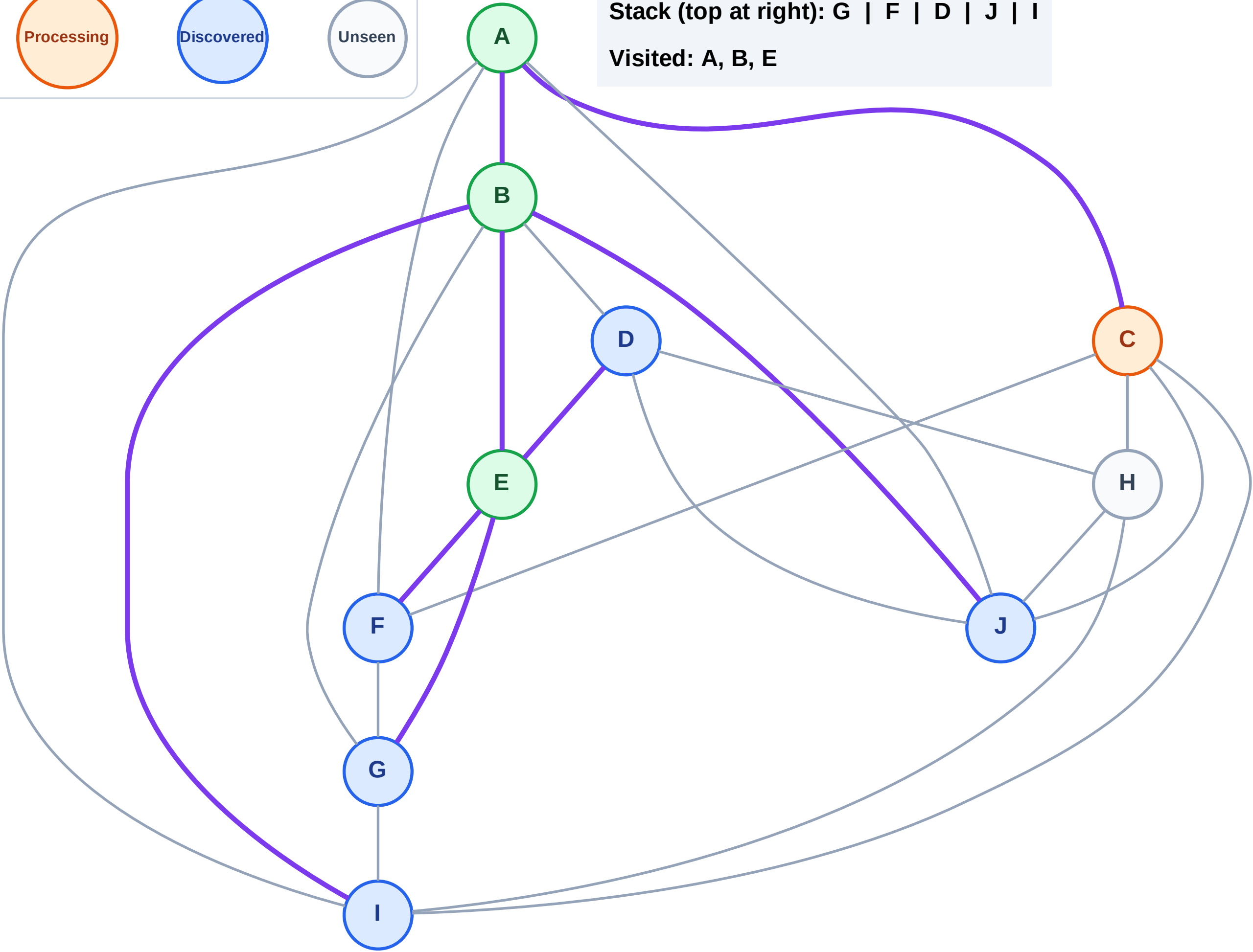
Processing

Discovered

Unseen

Stack (top at right): G | F | D | J | I

Visited: A, B, E



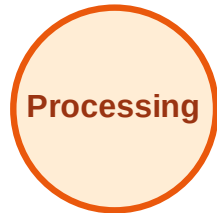
DFS Traversal

Step 9: Discover H from C

Legend



Complete



Processing



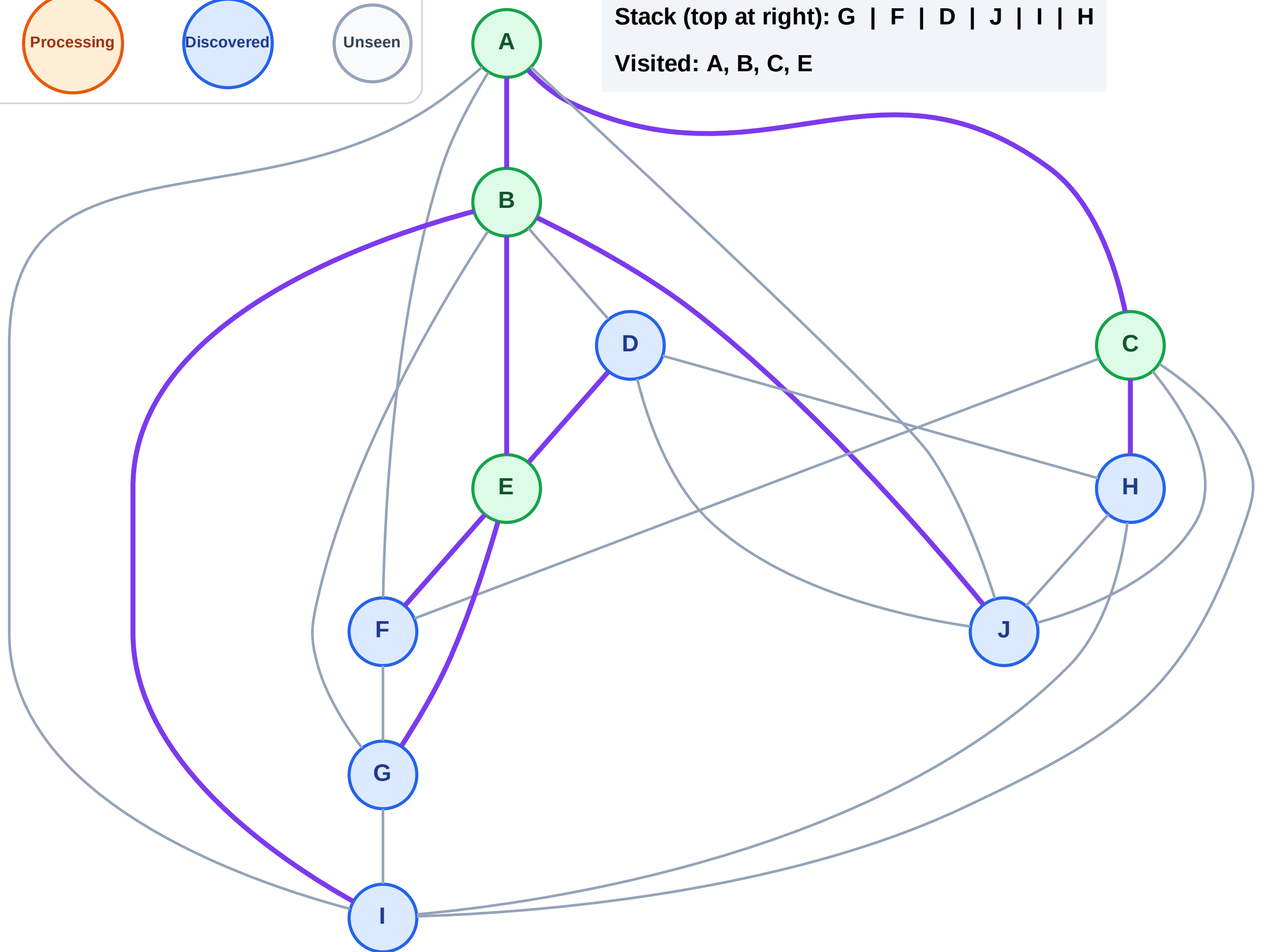
Discovered



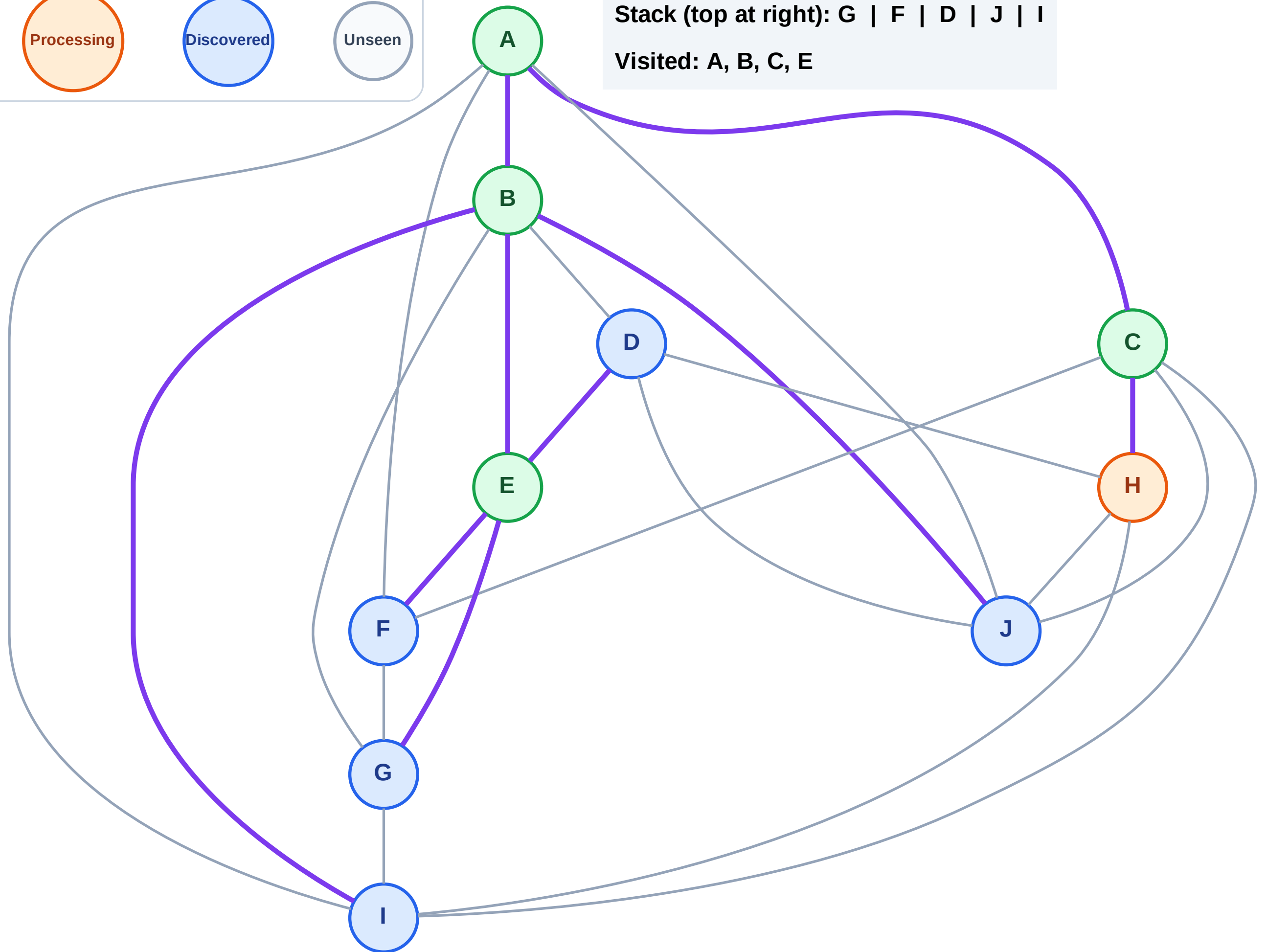
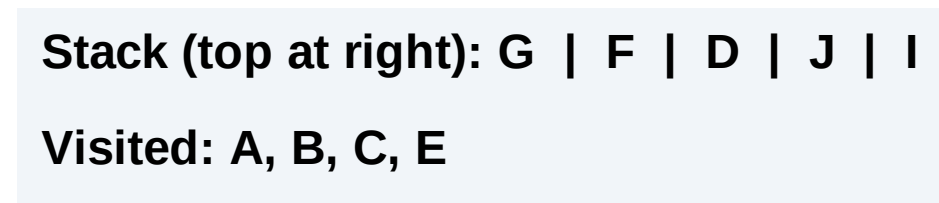
Unseen

Stack (top at right): G | F | D | J | I | H

Visited: A, B, C, E

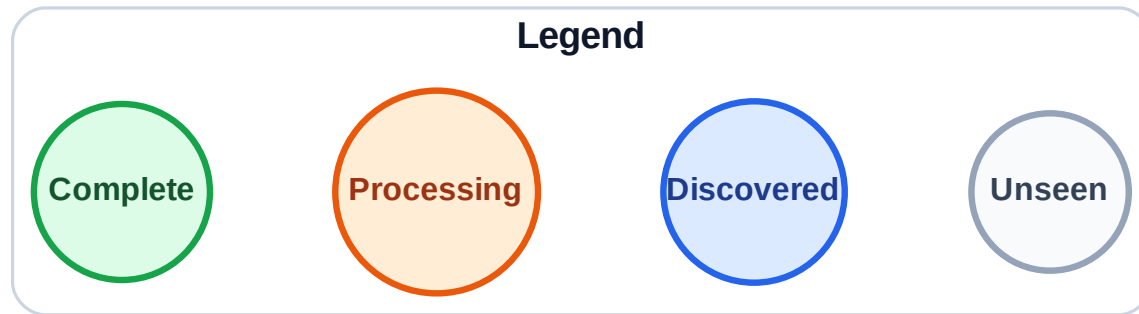


Step 10: Process node H



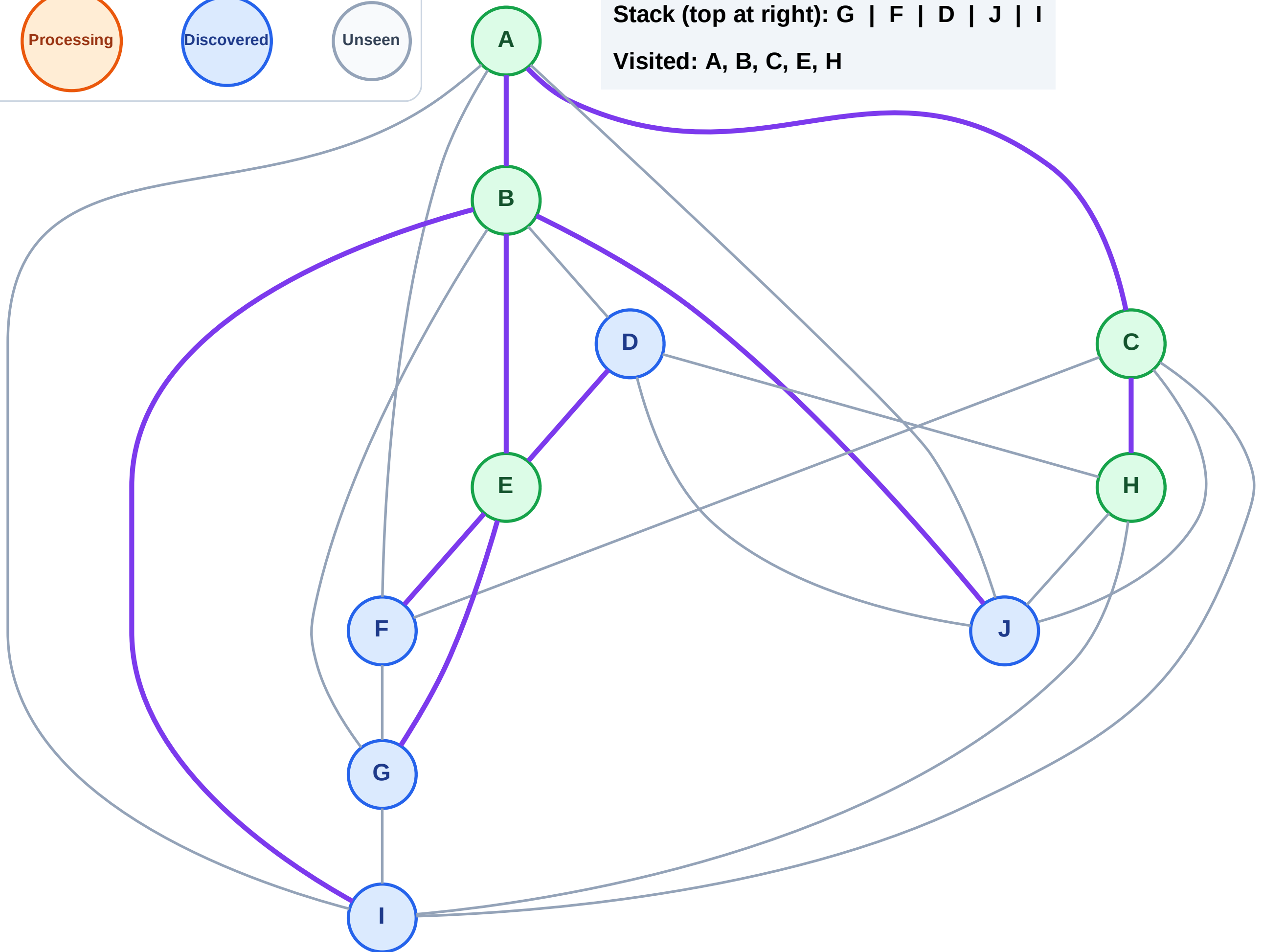
DFS Traversal

Step 11: No new neighbors from H



Stack (top at right): G | F | D | J | I

Visited: A, B, C, E, H



DFS Traversal

Step 12: Process node I

Legend

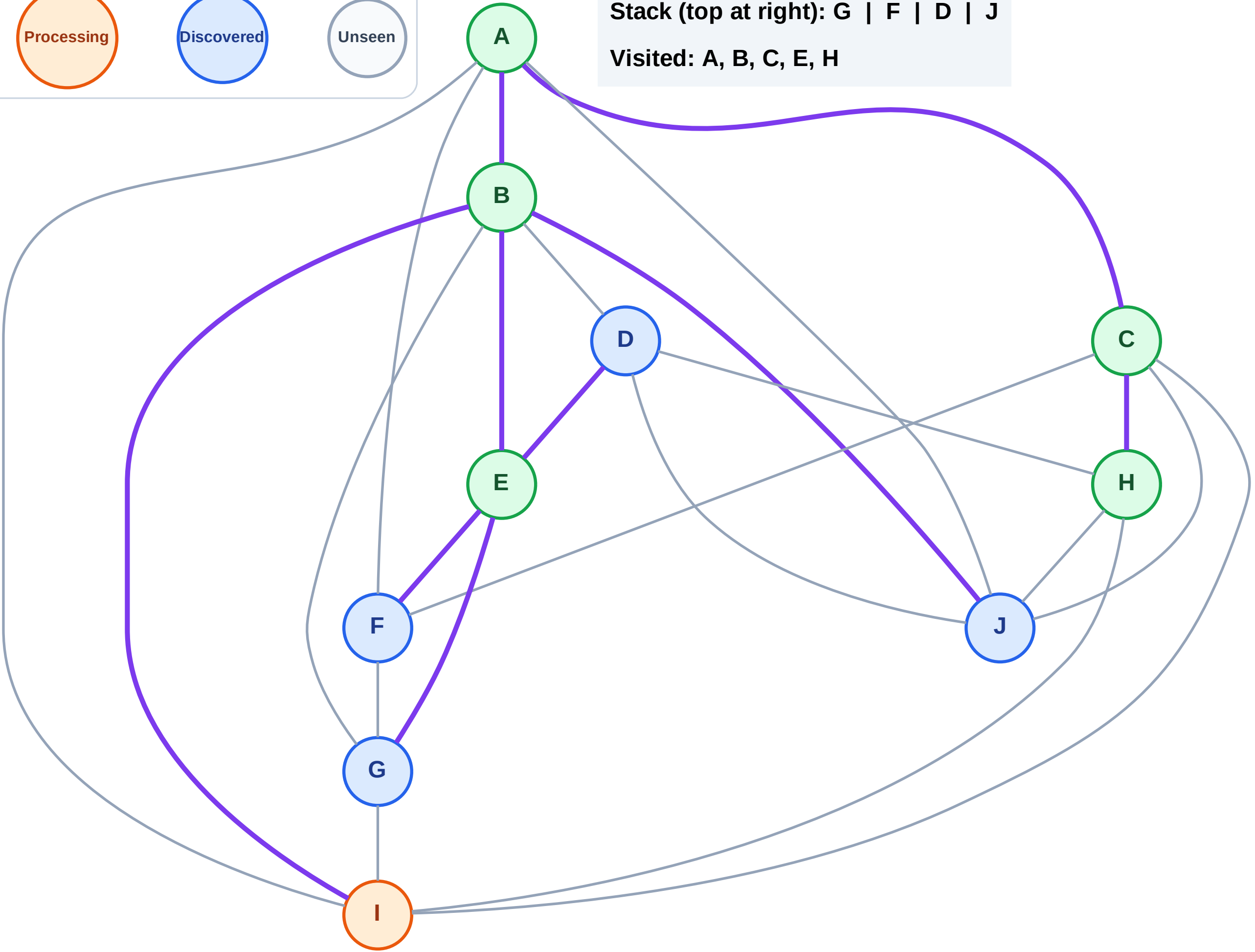
Complete

Processing

Discovered

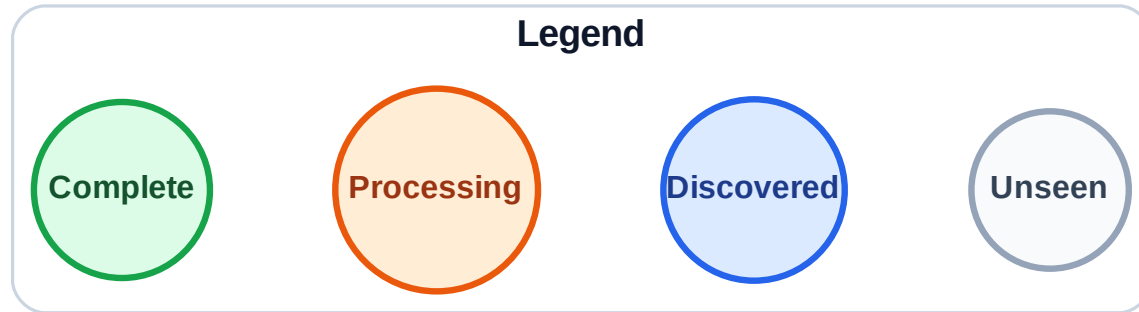
Unseen

Stack (top at right): G | F | D | J
Visited: A, B, C, E, H

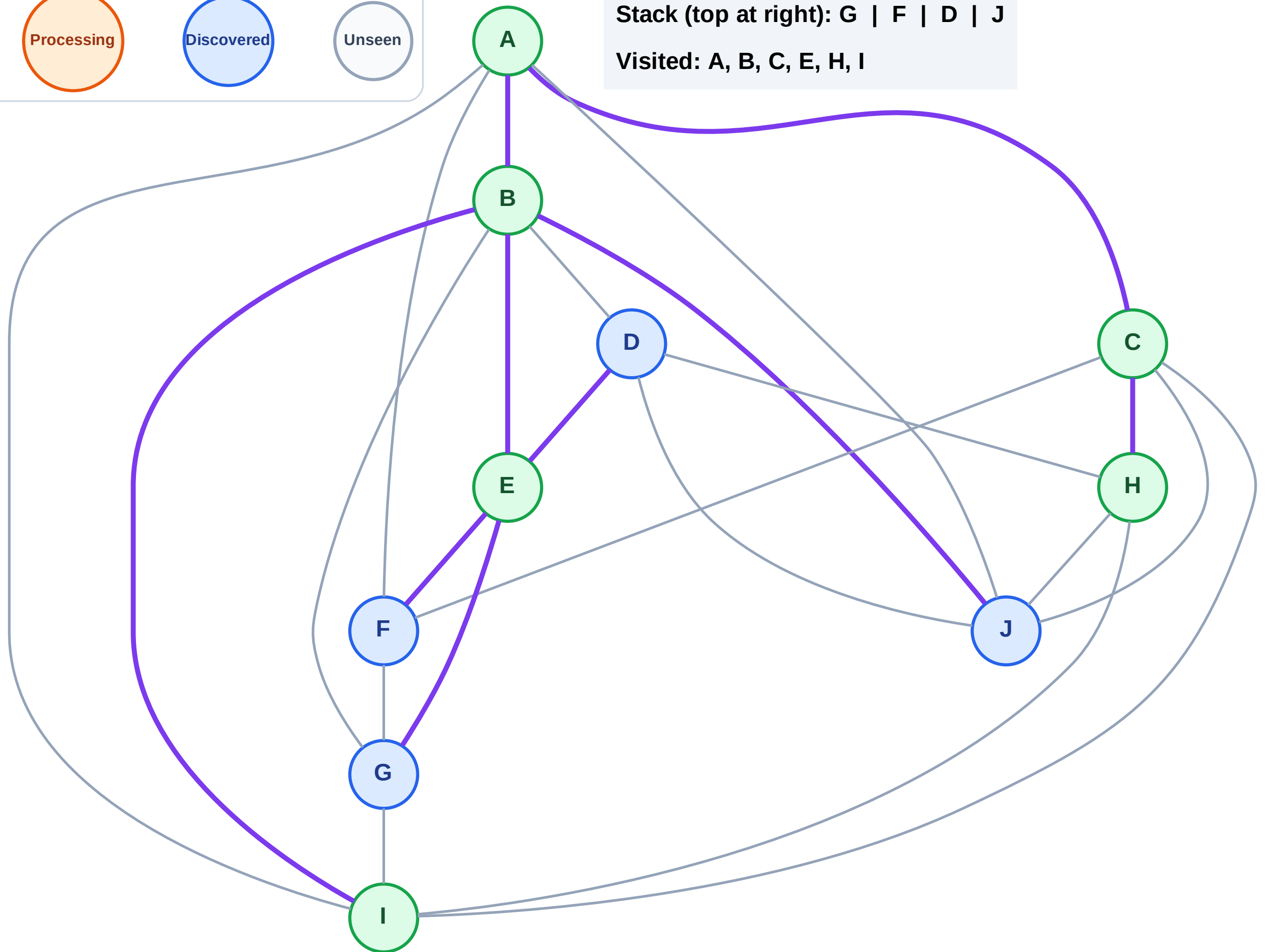


DFS Traversal

Step 13: No new neighbors from I



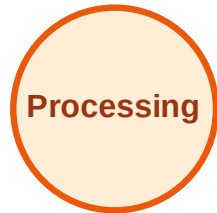
Stack (top at right): G | F | D | J
Visited: A, B, C, E, H, I



DFS Traversal

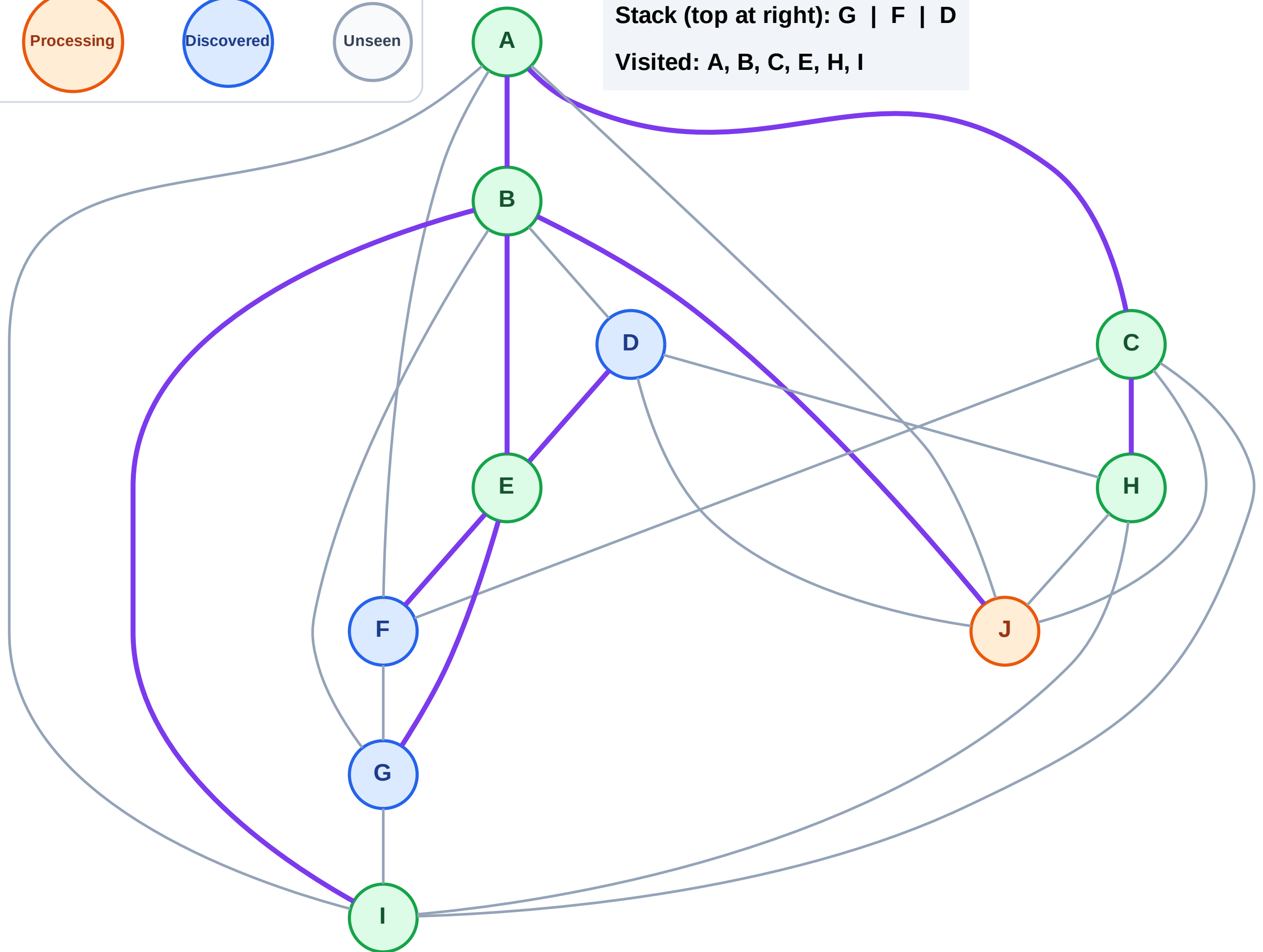
Step 14: Process node J

Legend



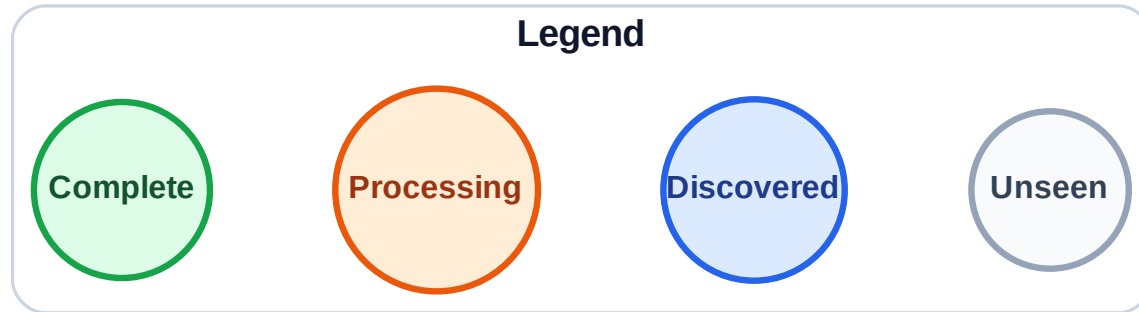
Stack (top at right): G | F | D

Visited: A, B, C, E, H, I

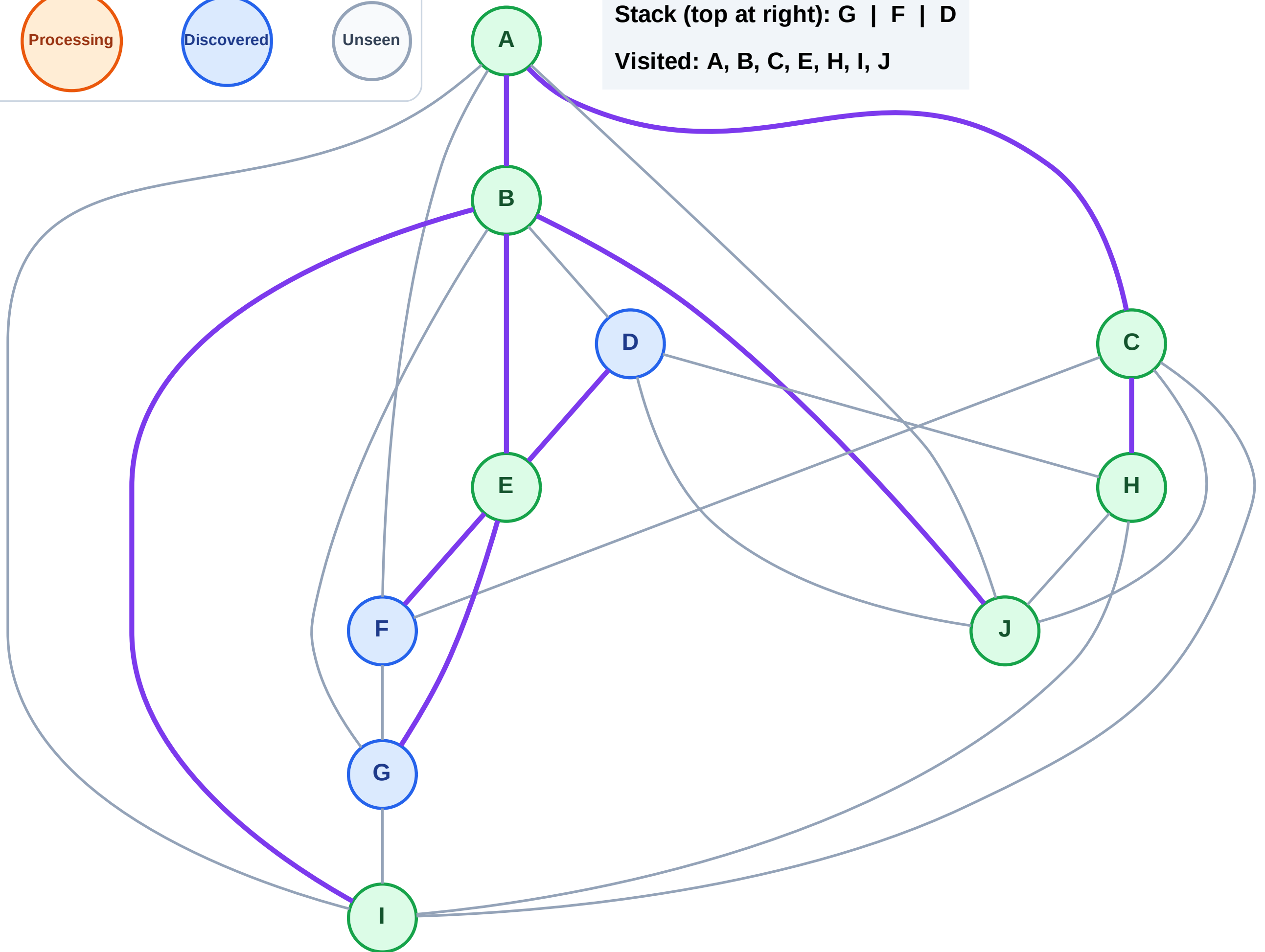


DFS Traversal

Step 15: No new neighbors from J



Stack (top at right): G | F | D
Visited: A, B, C, E, H, I, J



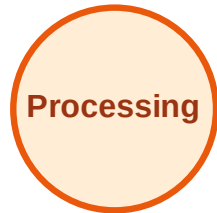
DFS Traversal

Step 16: Process node D

Legend



Complete



Processing



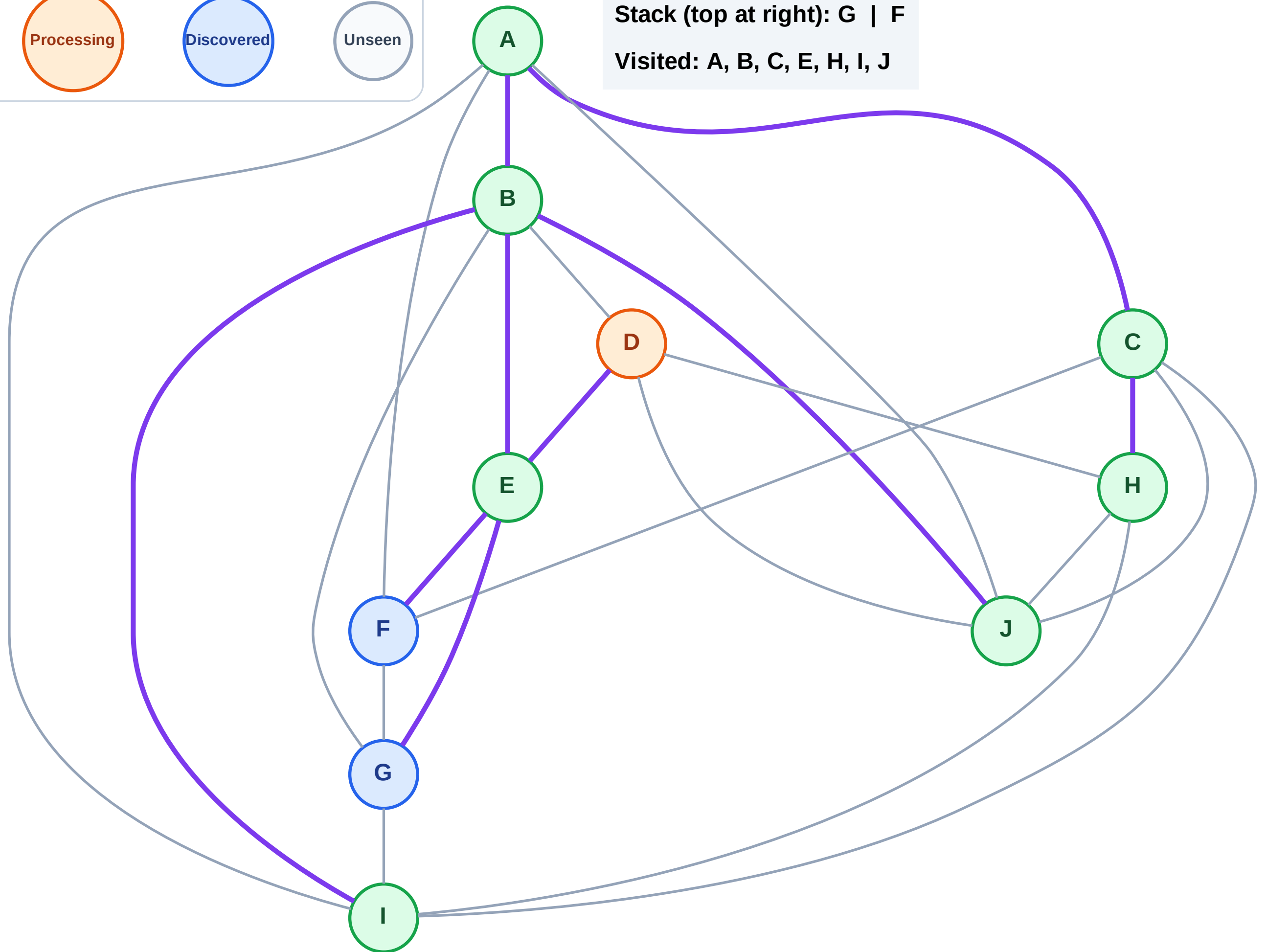
Discovered



Unseen

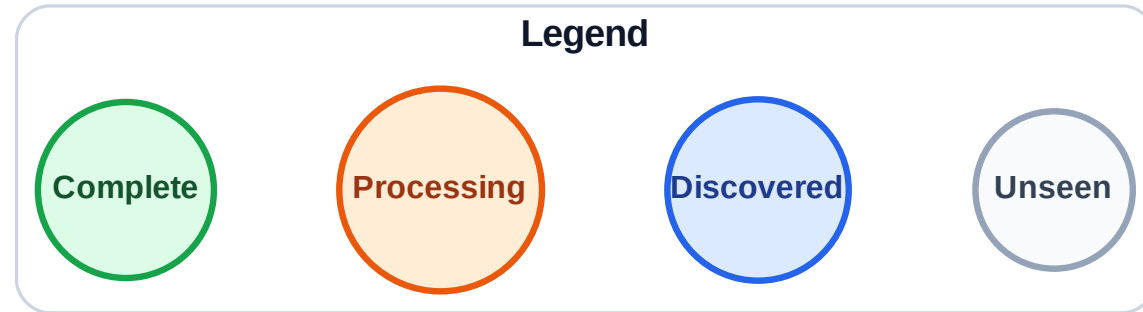
Stack (top at right): G | F

Visited: A, B, C, E, H, I, J

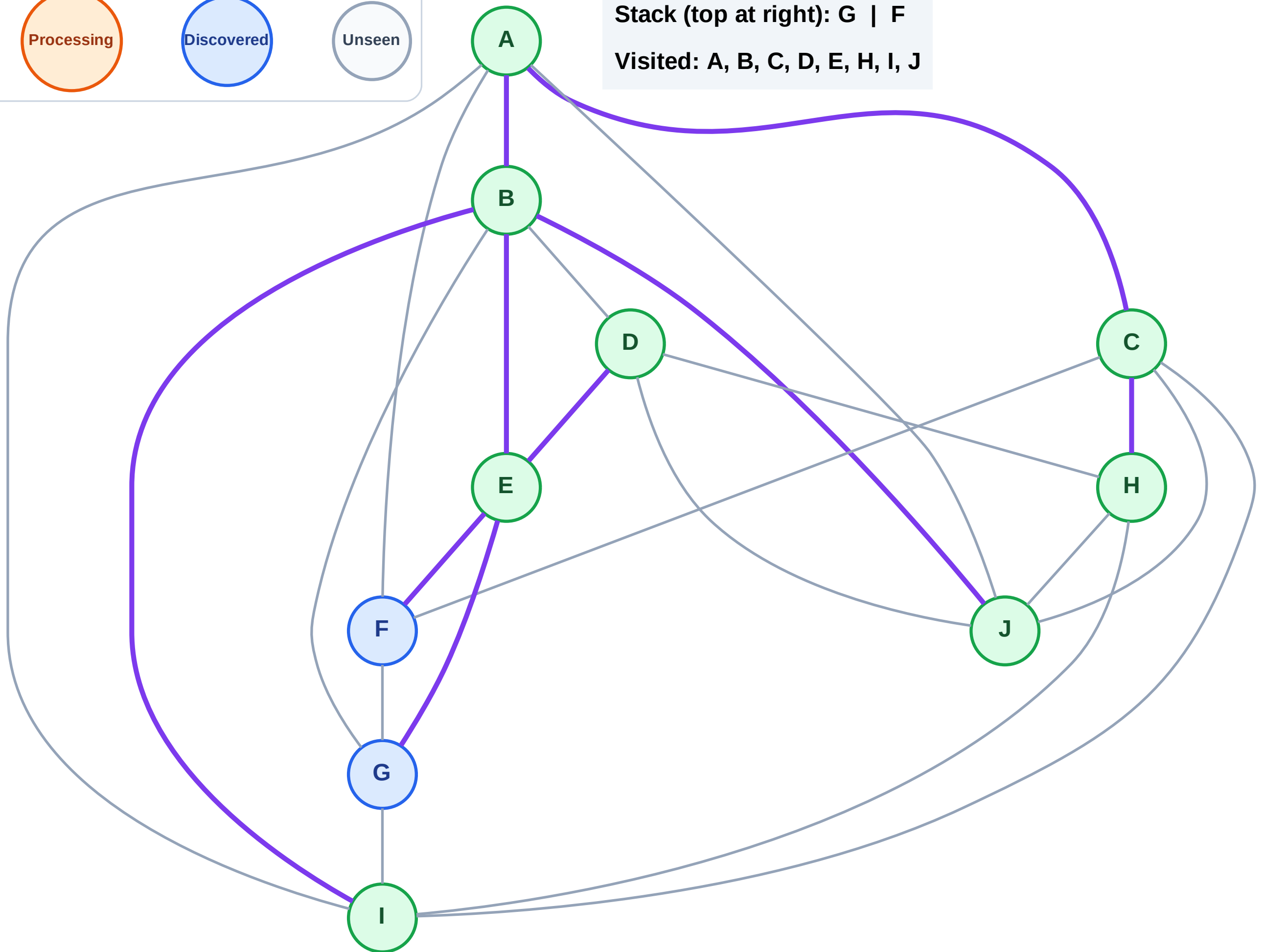


DFS Traversal

Step 17: No new neighbors from D



Stack (top at right): G | F
Visited: A, B, C, D, E, H, I, J



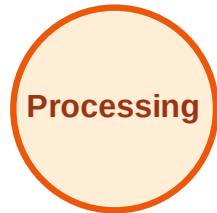
DFS Traversal

Step 18: Process node F

Legend



Complete



Processing



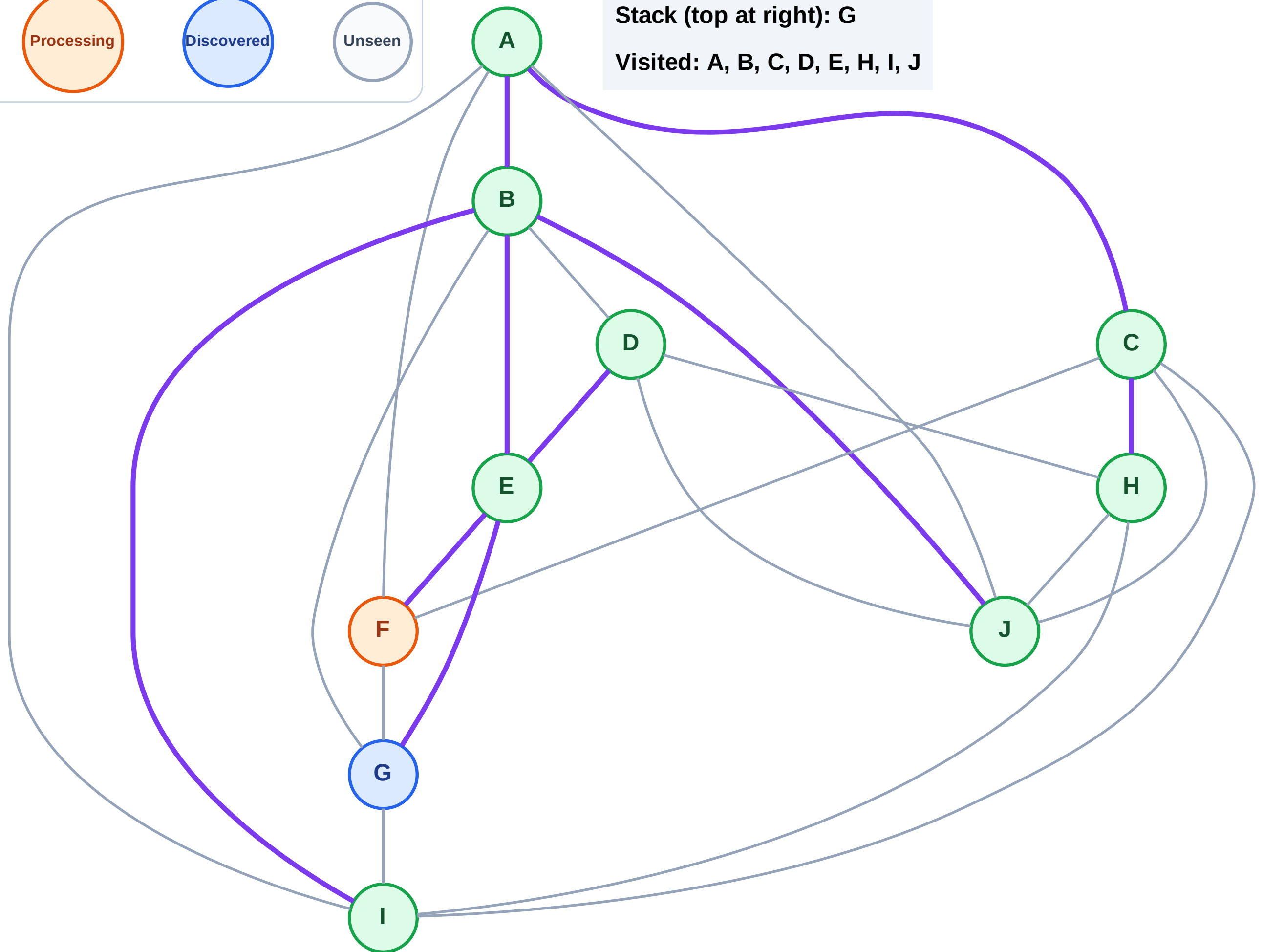
Discovered



Unseen

Stack (top at right): G

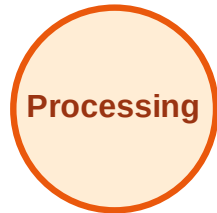
Visited: A, B, C, D, E, H, I, J



DFS Traversal

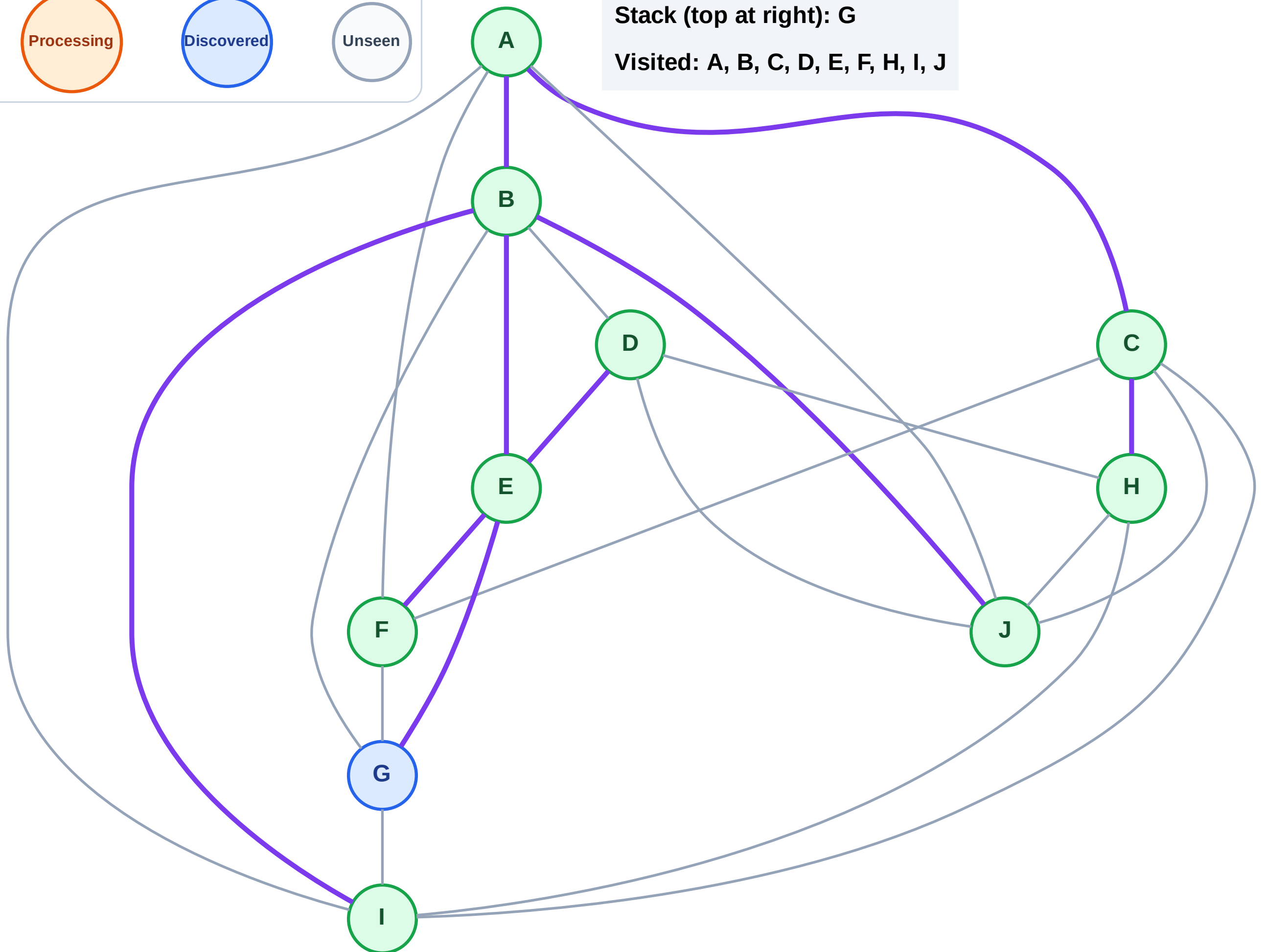
Step 19: No new neighbors from F

Legend



Stack (top at right): G

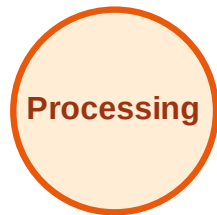
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

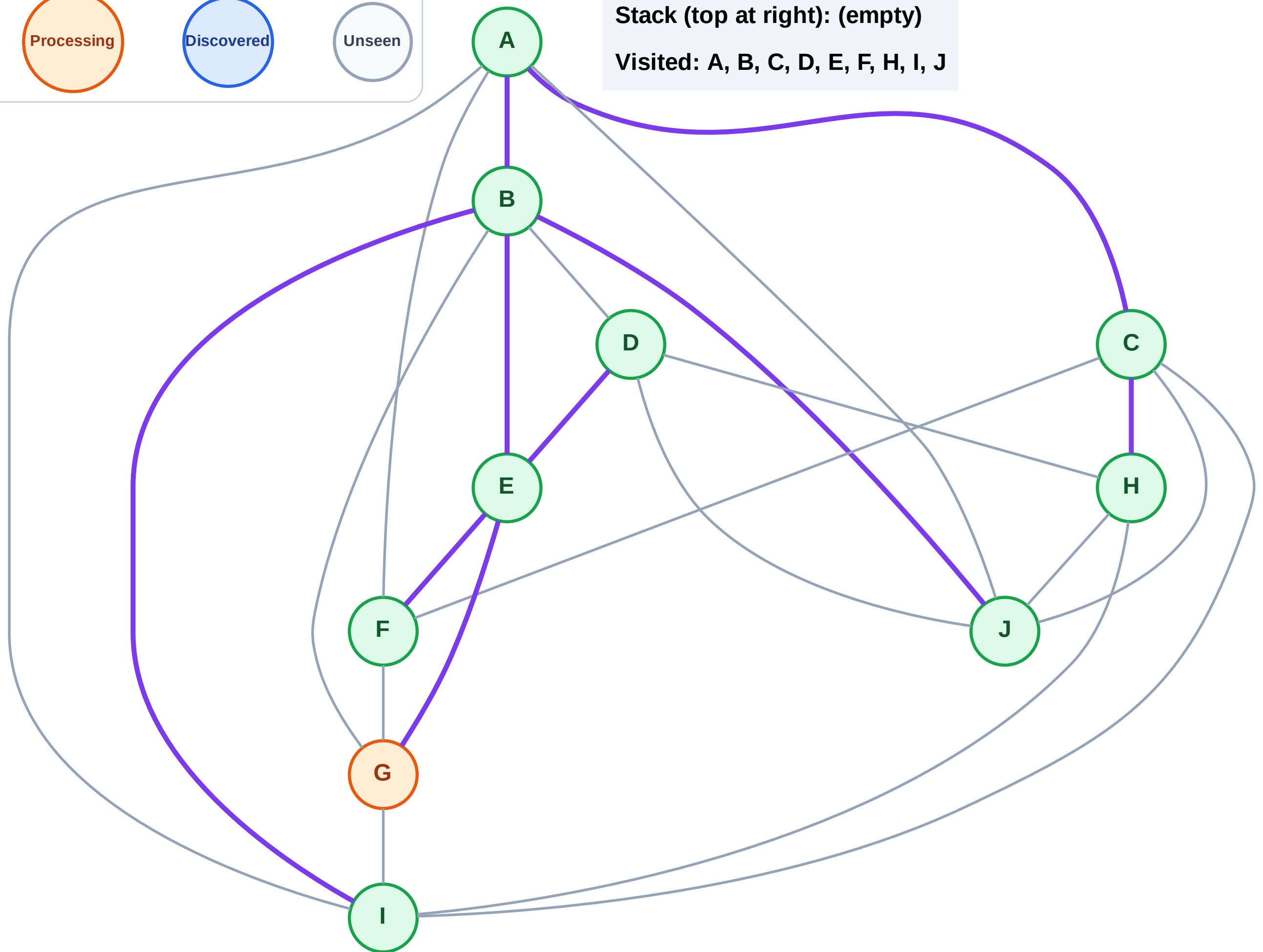
Step 20: Process node G

Legend



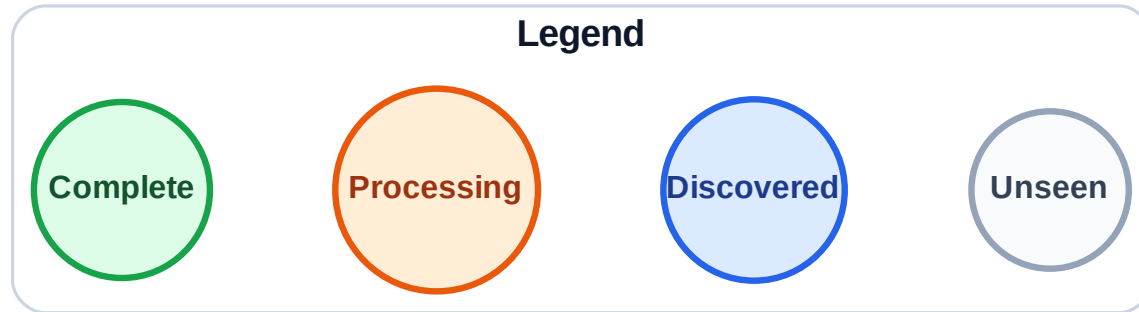
Stack (top at right): (empty)

Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

